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ADDIS ABABA INSTITUTE OF TECHNOLOGY

SCHOOL OF INFORMATION TECHNOLOGY AND ENGINEERING

Department of [Name]

Video Saliency Prediction with a Frozen Self-Supervised Video Encoder

A thesis submitted to the School of Information Technology and Engineering, Addis Ababa Institute of Technology, Addis Ababa University, in partial fulfillment of the requirements for the degree of Master of Science in [Program Name]

By

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Addis Ababa, Ethiopia

November 2026

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By

Full Name of the Student

Approval by Board of Examiners

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To . . .

ABSTRACT

Predicting where people look in videos supports applications from compression to content design and tests our understanding of human attention. Recent video saliency models draw their accuracy from large pretrained encoders, which they fine-tune on small eye-tracking datasets and run once for every output frame. This thesis asks whether a self-supervised video encoder trained by feature prediction, V-JEPA 2, is sufficient for the task without adapting it, and whether the error of the encoder's own pretrained predictor, a learned measure of surprise, indicates where people look.

The proposed model, SURGE-D, combines the frozen encoder with a decoder of 7.4 million parameters that reads four intermediate layers, rebuilds a multi-scale feature pyramid, decodes it with factorised spatio-temporal convolutions and predicts saliency maps for sixteen frames per forward pass. Every design choice was tested under a rule written before the experiment, all models were scored once per test set by an evaluator validated against reference code, and differences were assessed with a paired bootstrap over videos. The evaluation covers DHF1K, UCF Sports, Hollywood-2 with openly licensed gaze recordings, and six audio-visual datasets.

On the DHF1K validation set, a three-seed ensemble with the ViT-g encoder reached a correlation coefficient of 0.603 and a normalised scanpath saliency of 3.49, against 0.564 and 3.33 for SalFoM, the closest foundation-model approach, scored by the same evaluator. It was better on all six metrics, with bootstrap intervals that exclude zero, while needing about one sixth of SalFoM's encoder computation per output frame. After brief fine-tuning, the model was competitive with the best published results on UCF Sports and exceeded the best peer-reviewed results that could be verified on five of the six audio-visual datasets; it failed on conversations, where gaze follows the speaker. The predictor's error, by contrast, was anti-correlated with fixations and was beaten by simple frame differencing. Encoder pre-training and scale produced the only large gains, whereas low-rank adaptation, a learned centre prior, deep supervision, static-image co-training, longer clips, more training videos and audio fusion produced none that could be measured.

The thesis contributes an efficient frozen-encoder saliency model, a documented negative result on predictive surprise, controlled evidence on which design choices matter with a strong encoder, and a transparent evaluation protocol.

Keywords: video saliency prediction, eye fixations, self-supervised learning, joint-embedding predictive architecture, frozen encoder, predictive surprise

ACKNOWLEDGMENT

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LIST OF ACRONYMS

AAiT	Addis Ababa Institute of Technology
AAU	Addis Ababa University
AST	Audio Spectrogram Transformer
AUC	Area Under the ROC Curve
AUC-J	Area Under the ROC Curve, Judd variant
CC	Linear Correlation Coefficient
CI	Confidence Interval
CNN	Convolutional Neural Network
ConvLSTM	Convolutional Long Short-Term Memory
FiLM	Feature-wise Linear Modulation
FLOPs	Floating-Point Operations
FPN	Feature Pyramid Network
GELU	Gaussian Error Linear Unit
GN	Group Normalisation
GPU	Graphics Processing Unit
JEPA	Joint-Embedding Predictive Architecture
KL	Kullback–Leibler
KLD	Kullback–Leibler Divergence
LoRA	Low-Rank Adaptation
LSTM	Long Short-Term Memory
MAE	Masked Autoencoder
NSS	Normalised Scanpath Saliency
RNN	Recurrent Neural Network

ROC	Receiver Operating Characteristic
sAUC	Shuffled Area Under the ROC Curve
SIM	Similarity (histogram intersection)
SiTE	School of Information Technology and Engineering
UMT	Unmasked Teacher
V-JEPA	Video Joint-Embedding Predictive Architecture
ViT	Vision Transformer
VSP	Video Saliency Prediction

CHAPTER ONE

INTRODUCTION

1.1. Background

When people watch a video, they do not look everywhere. Their eyes move several times per second, and different observers tend to look at the same few regions: moving objects, faces and, in conversations, the person who is speaking [1], [2]. Where people look is shaped by the image itself, by the viewer's task and by strong regularities such as the preference for the centre of the screen [3], [4], [5]. *Video saliency prediction* is the task of predicting, for every frame of a video, the distribution of human fixations. Models of this kind support applications in which the allocation of attention matters, from compression and summarisation to the design and evaluation of visual content [6], and they are also scientific models of human attention whose successes and failures say something about what drives gaze.

The field has progressed in step with the representations it builds on. Early models computed saliency from the local contrast of hand-crafted features such as intensity, colour, orientation and motion [7]. The introduction of large eye-tracking benchmarks, most importantly DHF1K [8], [9], enabled deep networks to be trained for the task, and the accuracy of these networks has risen with each generation of their encoders: image networks with recurrent layers, three-dimensional convolutional networks pretrained for action recognition, video transformers and, most recently, video foundation models trained with self-supervision on large collections of video [10].

Self-supervised video models differ in what they learn to predict. One family, the joint-embedding predictive architectures, learns by predicting the *representations* of hidden parts of a video from the visible parts [11], [12]. Such models have two properties that make them interesting for saliency prediction. Their features are reported to work well without adapting the encoder to a new task [12], which suggests that a frozen encoder with a small task-specific decoder might suffice. And they contain a trained predictor, whose errors indicate what the model failed to anticipate, a learned analogue of the "surprise" that has been proposed as a driver of human attention [13].

1.2. Motivation

Three considerations motivated this work. The first is practical. SalFoM [10], the first video saliency model built on a video foundation encoder, which reported the best DHF1K test results among the methods it compared with, fine-tunes a 300-million-parameter encoder on a few hundred training videos and runs it once for every output frame. Fine-tuning so large a

model on so little data risks overfitting, which its authors offer as an explanation for its weaker results on a small dataset, and the per-frame design makes inference expensive. A frozen encoder with a decoder that predicts many frames at once would avoid both problems, if its features are good enough.

The second consideration is scientific. The idea that attention is drawn to the unexpected has strong support from computational models built on hand-crafted features [13], but it had not been tested with the learned predictor of a modern video model. If such a predictor's errors marked where people look, they would provide a principled attention cue learned without any gaze data; if they did not, understanding why would clarify what these models actually learn.

The third consideration is methodological. Published comparisons in video saliency rely mostly on numbers copied from other papers, computed with different evaluation code and sometimes on different data splits, and they rarely report uncertainty. Small reported gains are therefore hard to interpret. A study that re-scores its main baseline with the same validated evaluator, quantifies uncertainty and fixes its decision rules in advance can give answers that are more reliable, including negative answers.

1.3. Statement of the Problem

Recent video saliency models obtain their accuracy from large pretrained encoders, which they adapt to small saliency datasets, at a high computational cost per output frame. It is not known whether this adaptation is necessary, or whether the representations of a self-supervised video encoder trained by feature prediction are sufficient on their own, with only a light decoder trained on gaze data. Nor is it known whether the prediction errors of such an encoder's predictor, which offer a learned measure of surprise, carry information about where people look. Finally, because comparisons in the field are rarely made under a common evaluation protocol with uncertainty estimates, it is unclear which design choices, beyond the encoder, still matter once a strong encoder is used. This thesis addresses these questions with a frozen V-JEPA 2 encoder [14], a purpose-built decoder, and a pre-registered evaluation on nine eye-tracking test sets.

1.4. Research Questions

This study answers the following research questions:

RQ1. Can a frozen self-supervised video encoder of the V-JEPA family, combined with a lightweight spatio-temporal decoder, predict human fixations in video more accurately than the closest foundation-model approach, SalFoM, when both are evaluated under an identical protocol?

RQ2. Does the prediction error of the encoder's pretrained latent predictor ("predictive surprise") carry information about where people look, beyond that carried by simple

appearance and motion cues?

RQ3. Which design factors determine accuracy when the encoder is frozen: the pre-training and scale of the encoder, parameter-efficient adaptation, the learned centre prior, deep supervision, static-image co-training, temporal sampling, the amount of training data and ensembling?

RQ4. How well does the approach transfer to eye-tracking datasets with other content and viewing tasks, and does adding the soundtrack improve it on audio-visual datasets?

1.5. Objectives

1.5.1. General Objective

The general objective of this study is to design, implement and rigorously evaluate a video saliency prediction model built on a frozen self-supervised video encoder, and to test whether the encoder's own prediction error is a useful cue for human attention.

1.5.2. Specific Objectives

To achieve the general objective, the study will:

- review the literature on visual attention, video saliency prediction and self-supervised video representation learning;
- prepare the eye-tracking datasets and an evaluation protocol whose metric implementation is validated against reference code, including a new ground truth for Hollywood-2 from openly licensed gaze recordings;
- design and implement a decoder that turns the intermediate features of a frozen V-JEPA 2 encoder into saliency maps for many frames per forward pass;
- test the predictive surprise of the V-JEPA 2 predictor as a fixation cue against simple reference cues;
- compare encoders and design choices under decision rules fixed before each experiment;
- evaluate the final models on DHF1K and eight further test sets, with paired statistical comparisons where the data allow; and
- measure the computational cost of the approach.

1.6. Research Methodology

1.6.1. Literature Review

The literature was searched in publisher indexes and through the reference lists of key papers, covering visual attention, image and video saliency prediction, video representation learning

and evaluation methodology. Only peer-reviewed publications are cited. The venue and bibliographic details of every reference were verified against the publishers' records or, for proceedings without digital object identifiers, against the indexes of those proceedings; works available only as preprints were excluded.

1.6.2. Data Collection

No new gaze data were recorded. The study uses public eye-tracking datasets: DHF1K, UCF Sports, Hollywood-2 with the gaze recordings of Vig et al., the six audio-visual datasets of the STAViS benchmark, and SALICON for one ablation (Section 4.3).

1.6.3. Tools and Techniques

The model was implemented in Python with PyTorch [15]; the pretrained encoders were loaded with the Transformers library [16]. All experiments ran on a single workstation with one NVIDIA RTX A5000 graphics card (Section 5.1).

1.6.4. Evaluation

Models were evaluated with six standard saliency metrics [17], computed by an evaluator validated against reference code. Differences between models were tested with a paired bootstrap over videos, and every design decision was governed by a rule written before the corresponding experiment (Section 4.1).

1.7. Scope and Limitations

Scope. The study covers offline (non-causal) saliency prediction for video, in which the map of a frame may depend on the frames around it. The main model is visual; the soundtrack is examined in one separate experiment. The encoders are frozen V-JEPA 2 models in two sizes; V-JEPA 2.1 and UMT-L are included in the encoder comparison. Evaluation covers nine test sets: DHF1K, UCF Sports, Hollywood-2 with the Vig gaze data, and six audio-visual datasets.

Limitations. The DHF1K test-set annotations are withheld by the dataset authors; predictions for the test set were prepared, but their scores were not available when the thesis was written, so DHF1K results are reported on the official validation set. The standard Hollywood-2 ground truth could not be obtained, so the Hollywood-2 results use other gaze data and are not comparable with published numbers. Full fine-tuning of the encoder, other foundation encoders and causal (streaming) prediction were outside the scope of the study, and the component ablations used a single training seed because of the available computing resources.

1.8. Contributions

The main contributions of this thesis are:

- **SURGE-D**, a video saliency model that combines a frozen V-JEPA 2 encoder with a light feature-pyramid decoder of factorised spatio-temporal convolutions and predicts maps for sixteen frames per forward pass. With 7.4 million trained parameters, its three-seed ViT-g ensemble reaches a correlation coefficient of 0.603 on the DHF1K validation set, against 0.564 for SalFoM scored by the same evaluator, and is better on all six metrics with paired bootstrap intervals that exclude zero. Counting the encoder alone, SalFoM's per-frame design requires about six times more computation per output frame.
- **A negative result on predictive surprise**: the error of the pretrained V-JEPA 2 predictor does not indicate where people look; it is beaten by frame difference and is even anti-correlated with fixations, because it mainly reflects the atypicality of regions rather than violated expectations.
- **Controlled evidence on what matters** once the encoder is frozen: the encoder's pre-training and scale produce the only large gains, seed ensembles a small one, while low-rank adaptation, the learned centre prior, deep supervision, static-image co-training, longer clips, more training videos and a simple audio fusion produce none that could be measured.
- **A transparent evaluation** across nine test sets with a validated evaluator, pre-registered decision rules, paired statistics and a new, openly reproducible Hollywood-2 ground truth from the gaze recordings of Vig et al., with the code, splits and decision log provided with the thesis.

1.9. Organization of the Thesis

The rest of this thesis is organized as follows. Chapter 2 reviews the background concepts and techniques. Chapter 3 discusses related work and identifies the research gap. Chapter 4 presents the proposed approach and the evaluation protocol. Chapter 5 describes the experiments and discusses the results. Chapter 6 concludes the thesis and outlines future work.

CHAPTER TWO

LITERATURE REVIEW

This chapter introduces the concepts on which the thesis builds. It begins with the study of visual attention and the notion of a saliency map (Section 2.1), follows the development of computational saliency models from hand-crafted features to deep networks trained on large datasets (Section 2.2), and then reviews how video representations are learned, with emphasis on the self-supervised video encoders that this thesis uses (Section 2.3). The last two sections describe the eye-tracking datasets and the evaluation metrics of the field (Sections 2.4 and 2.5). Specific video saliency models are reviewed in Chapter 3.

2.1. Visual Attention and Saliency

2.1.1. Bottom-Up and Top-Down Attention

The human visual system cannot process the whole visual field at full resolution. High acuity is limited to the fovea, so the eyes move several times per second to bring regions of interest into central vision, and the sequence of fixations reveals what the observer attends to. Two broad influences on this selection are usually distinguished: *bottom-up* factors, which depend on the properties of the image, such as local contrast in colour, orientation or motion; and *top-down* factors, which depend on the observer's goals, knowledge and task [3], [4].

The feature-integration theory of Treisman and Gelade [18] proposed that simple features are registered in parallel across the visual field, whereas attention is required to bind them into objects. Building on this view, Koch and Ullman [19] introduced the *saliency map*: a topographic map that combines the conspicuity of locations across feature dimensions, from which a winner-take-all mechanism selects the next location to attend, while inhibition of return prevents attention from returning immediately to the same place. Itti, Koch and Niebur [7] turned this idea into an influential computational model, in which centre–surround differences of intensity, colour and orientation at several scales are combined into a single saliency map. Their model, and the many that followed it, defined saliency in terms of local feature contrast [4]. Graph-based visual saliency, for example, computes activation maps from feature maps and normalises them through Markov chains [20], while the SUN model defines bottom-up saliency as the self-information of features relative to the statistics of natural images [21].

Top-down influences are harder to model because they depend on what the observer is trying to do. Judd et al. [22] showed that a model learned from eye-tracking data, combining low-level features with higher-level detectors for faces and people and with a centre prior,

predicted fixations better than purely bottom-up models. This shift, from designing saliency features to learning them from human data, set the direction for the deep models that followed.

2.1.2. Attention in Dynamic Scenes

Video adds time, and with it motion, events and narrative. In dynamic scenes, gaze is strongly attracted by motion: in the DIEM recordings, the clustering of gaze across viewers was predicted by motion [1]. Carmi and Itti [23] distinguished features that cause attentional selection from features that merely correlate with it, using video clips with abrupt scene cuts, after which the influence of memory and expectation on gaze is briefly disrupted.

Two further biases are prominent in video. The first is the *centre bias*. Observers tend to fixate the centre of the screen, and Tatler [24] showed that this bias cannot be explained by motor tendencies or by the distribution of image features alone. For dynamic scenes, Tseng et al. [5] found that the centre bias correlates strongly with *photographer bias*, the tendency to frame interesting content centrally, and is influenced by viewing strategy at scene onsets. The second is the strong attraction of *faces*, particularly of people who speak: in recordings of conversations, observers look mostly at faces, and especially at talking faces, and the original soundtrack makes them follow the turn-taking of the speakers more closely [2]. Sound thus influences gaze even when it carries no spatial information [25].

2.1.3. Surprise and Prediction

A distinct account holds that attention is drawn to events that violate expectations. Itti and Baldi [13] defined *Bayesian surprise* as the difference between an observer's posterior and prior beliefs after an observation. In their experiments, 72 % of the gaze shifts of observers watching television were directed towards locations more surprising than average, rising to 84 % for regions selected by all observers. Their model computed surprise from simple simulated early visual neurons, and earlier work had used it to detect surprising events in video [26]. The account connects to predictive-coding theories of the visual cortex, in which neural responses signal the error between the input and a prediction formed from context [27]. Whether the prediction errors of modern learned video models play a similar role is one of the questions examined in this thesis.

2.2. Deep Saliency Prediction for Images

The arrival of large eye-tracking and attention datasets enabled saliency models to be learned end to end. SALICON [28] collected mouse-contingent attention data, a scalable proxy for eye tracking, on images from MS COCO, and thereby provided a dataset large enough to train deep networks for saliency. Pan et al. [29] trained convolutional networks directly on such data, while Huang et al. [30] fine-tuned networks pretrained for object recognition and argued

that their semantic features close much of the gap between low-level saliency and human fixations. Cornia et al. [31] refined the prediction iteratively with an attentive convolutional LSTM and learned a set of Gaussian prior maps to model the centre bias, a design that recurs in later models, including the one proposed in this thesis.

The DeepGaze line of work clarified *why* deep models succeed. Kümmerer et al. [32] separated the contributions of low- and high-level features to fixation prediction, and later showed that transfer learning from pretrained backbones had been the key driver of progress, with a twist: replacing the VGG19 backbone of DeepGaze II by ResNet50 features raised performance considerably, but better ImageNet classifiers, such as EfficientNet, brought no further gain [33]. The quality of the pretrained representation for saliency is therefore not simply its quality for classification. Borji [6] reviewed the deep-learning era of saliency prediction for images and videos, and identified the remaining gap to human performance, the failure cases of current models and the difficulty of fair model comparison as open issues.

2.3. Learning Representations of Video

2.3.1. Spatio-Temporal Convolutional Networks

Early deep video models extended image convolutions into time. C3D learned spatio-temporal features with three-dimensional convolutions on video clips [34]. Carreira and Zisserman inflated two-dimensional networks pretrained on ImageNet into three dimensions (I3D) and introduced Kinetics, a large dataset of human actions that became the standard source of video pre-training [35]. Xie et al. [36] showed that many of the costly three-dimensional convolutions can be replaced by separable spatial and temporal ones, producing the S3D network that many video saliency models of the following years use as their encoder. Tran et al. [37] analysed such factorisations systematically; their R(2+1)D block, a spatial convolution followed by a temporal one, is the basis of the decoder blocks in this thesis.

2.3.2. Video Transformers and Dense Prediction

The Transformer [38] replaced recurrence and convolution by self-attention, which relates every element of a sequence to every other. The Vision Transformer (ViT) [39] applied it to images by splitting them into patches of 16×16 pixels that are embedded as tokens. For video, ViViT [40] and TimeSformer [41] extract spatio-temporal tokens and factorise attention over space and time to cope with the long token sequences, while Video Swin [42] restricts attention to local windows, reintroducing an inductive bias of locality.

A plain ViT produces features at a single resolution, whereas dense prediction tasks need several. Feature pyramid networks [43] build multi-scale representations from convolutional backbones with a top-down pathway and lateral connections. For transformers, DPT [44] assembles tokens from several stages into image-like maps at different resolutions and fuses

them with a convolutional decoder, and Li et al. [45] showed that a simple pyramid built by up- and down-sampling the features of a plain ViT suffices for object detection. The decoder of this thesis follows these designs (Section 4.4).

2.3.3. Self-Supervised Video Encoders

Supervised pre-training ties a representation to a fixed set of labels. Self-supervised learning instead derives the training signal from the data itself, which allows much larger and more varied training sets. Masked autoencoders (MAE) [46] mask a large fraction of image patches and train the network to reconstruct their pixels; VideoMAE [47] applied the same principle to video with very high masking ratios. UMT [48] instead aligns the unmasked tokens of a video student with the features of an image foundation model, which speeds up training and gives semantically rich features; its ViT-L is the encoder of SalFoM (Section 3.4).

Joint-embedding predictive architectures (JEPA) take a third route. Rather than reconstructing pixels, they predict the *representations* of masked parts of the input. I-JEPA [11] predicts the representations of several target blocks of an image from a single context block, and V-JEPA [12] extends this to video. With x the visible (context) part of a clip, y the full clip, E_θ the encoder, $\bar{E}$ an exponential-moving-average copy of it, P_ϕ the predictor and $\mathcal{M}$ the set of masked tokens, the objective can be written as [14]

$$\mathcal{L}_{\text{JEPA}} = \frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} \left\| P_\phi\big(E_\theta(x), \Delta_y\big)_i - \text{sg}\big(\bar{E}(y)_i\big) \right\|_1, \tag{2.1}$$

where Δ_y are learnable tokens that mark the positions to be predicted and sg stops the gradient. Because the target is a representation rather than pixels, the encoder can ignore unpredictable detail. Bardes et al. [12] report that the resulting features perform well on both appearance- and motion-oriented tasks without adapting the encoder, that is, with a frozen backbone. A related observation for images is that self-supervised features can serve many tasks without fine-tuning [49].

Since the loss in Equation (2.1) is applied only to the masked tokens, the visible tokens receive no direct training signal. Mur-Labadia et al. [14] found that the features of the V-JEPA 2 models consequently carry little local structure (22.2 mIoU in linear-probe semantic segmentation on ADE20K), and proposed V-JEPA 2.1, which adds a loss on the visible tokens and self-supervision at intermediate layers to obtain denser features. Both model families are used in this thesis.

A JEPA model also contains its predictor P_ϕ. After training, the predictor's error on a new video, the difference between what it predicted for a region and what the encoder actually computed there, is a learned measure of how unexpected that region is. This is the quantity examined as a possible attention cue in Section 5.4.

2.3.4. Adapting Pretrained Models

A pretrained encoder can be used frozen, with only a task-specific head trained on top, or adapted to the task. Full fine-tuning updates all parameters and can overfit small datasets. Parameter-efficient methods adapt a model with few trainable parameters: adapters insert small bottleneck layers into a frozen network [50], and LoRA [51] adds trainable low-rank updates to the weight matrices of a frozen transformer, matching full fine-tuning on language tasks while training a small fraction of the parameters. Conditioning information, such as an audio signal, can be injected into a network through feature-wise linear modulation (FiLM), which scales and shifts intermediate features as a function of the conditioning input [52]. Finally, averaging the predictions of several independently trained models, a deep ensemble, is a simple way of improving accuracy and calibration [53]. Each of these techniques is evaluated in this thesis.

2.4. Eye-Tracking Datasets for Video

Table 2.1 lists the video eye-tracking datasets relevant to this thesis. They differ in size, content and, importantly, viewing task.

Table 2.1: Video eye-tracking datasets relevant to this thesis

Dataset	Videos	Viewers	Content and task
DHF1K [8], [9]	1,000	17	varied YouTube videos, free viewing; test labels withheld
LEDOV [54]	538	32	varied videos, free viewing
Hollywood-2 [55], [56]	1,707	19	movie clips; most viewers performed action recognition
UCF Sports [56], [57]	150	19	sports videos; same protocol as Hollywood-2
Hollywood-2, Vig et al. [58]	1,707	4 + 1 pooled	independent recording during action recognition
DIEM [1]	84	42	advertisements, trailers, documentaries; audio on
Coutrot1 [2], [25]	60	72	four visual categories; with and without sound
Coutrot2 [2]	15	40	four people in meetings
ETMD [59], [60]	12	10	clips from Hollywood movies
AVAD [61]	45	16	short audio-visual scenes
SumMe [60], [62]	25	10	user-made videos

DHF1K [8], [9] is the largest free-viewing benchmark: 1,000 videos annotated by 17 observers, split into 600 training, 100 validation and 300 test videos whose annotations are withheld and evaluated by the dataset authors on request. Hollywood-2 and UCF Sports are action recognition datasets [55], [57] for which Mathe and Sminchisescu [56] recorded the gaze of 19 observers, most of whom were asked to recognise the action; their ground truth is therefore task-driven and concentrated on the actors. Vig et al. [58] independently recorded gaze on Hollywood-2 during action recognition and released the raw recordings openly. The six audio-visual datasets of the lower part of Table 2.1 were assembled into a

common benchmark by Tsiami et al. [63]; the viewer counts in the table follow their description and that of later work [64]. Fixation maps in all these datasets are converted into continuous density maps by Gaussian smoothing, and the binary fixation maps are kept for location-based metrics.

The differences in task are central to interpreting results across datasets. Free viewing (DHF1K, DIEM) captures where people look spontaneously; the action recognition task (Hollywood-2, UCF Sports) concentrates gaze on the actors and their movements; and in conversations (Coutrot2) gaze follows the speaker. A model trained on one condition need not transfer to another, which is why results on these datasets are usually obtained after fine-tuning on each.

2.5. Evaluation Metrics

No single metric captures all aspects of a saliency map. Bylinskii et al. [17] analysed the common metrics and grouped them into *location-based* metrics, which evaluate the predicted map at the fixated locations, and *distribution-based* metrics, which compare the predicted map with the continuous ground-truth density. The location-based family includes the normalised scanpath saliency (NSS) [65], the mean of the standardised prediction at fixated pixels, and the area under the ROC curve in the variant of Judd et al. (AUC-J) [22], which separates fixated from non-fixated pixels. The shuffled AUC (sAUC) draws its negatives from the fixations made on other images or videos, so that a model is not rewarded merely for predicting the centre bias [21], [66]. The distribution-based family includes the linear correlation coefficient (CC), the histogram intersection (SIM) and the Kullback–Leibler divergence (KLD). Le Meur and Baccino [67] discuss the strengths and weaknesses of these comparisons, and Section 4.8 gives the exact definitions used in this thesis.

The metrics respond differently to the same map [17]. KLD heavily penalises predictions that assign almost no probability to regions that observers fixate; SIM measures the overlap of two distributions and is sensitive to ground-truth mass that the prediction misses; the AUC metrics depend only on the ranking of saliency values; and sAUC penalises centre bias. Kümmerer et al. [68] proposed to compare models in terms of information gain over a baseline, which unifies several metrics in a probabilistic framework, and later showed that the ranking of models depends on how a probabilistic prediction is turned into a map for each metric. They argued that each metric should be given the map that is optimal for it under the model's prediction [69]; in practice, video saliency benchmarks score one map per model with all metrics. In video saliency practice, the metrics are computed per frame, averaged over the frames of each video and then over videos, usually with evaluation code derived from the MIT saliency benchmark [17]; small differences between implementations make numbers from different papers only approximately comparable.

2.6. Summary

Visual attention is shaped by image features, by the observer's task and by strong regularities such as the centre bias and the attraction of faces, and in video additionally by motion, events and sound. Computational models moved from hand-crafted feature contrast to deep networks whose success rests largely on pretrained representations. Video representation learning followed a parallel path from three-dimensional convolutions to transformers and, most recently, to self-supervised encoders trained on large video collections; among these, the joint-embedding predictive models are distinctive in learning by predicting representations, and in containing a predictor whose errors could indicate what is unexpected. The datasets and metrics of the field are diverse in task and in what they reward, which makes careful and consistent evaluation essential. The next chapter reviews how these ingredients have been combined into video saliency models.

CHAPTER THREE

RELATED WORK

Deep video saliency models differ mainly in two respects: how they represent the temporal dimension of a video, and what their encoder has been trained on before it meets eye-tracking data. This chapter follows that development, from recurrent models built on image features (Section 3.1), through three-dimensional convolutional encoders (Section 3.2) and transformers (Section 3.3), to models that build on video foundation encoders (Section 3.4). Audio-visual models, which add the soundtrack, are reviewed separately (Section 3.5), as is the small body of work that treats prediction error as a cue for attention (Section 3.6). The chapter closes with a summary and the research gap that this thesis addresses (Section 3.7).

3.1. Recurrent and Two-Stream Models

The first deep video saliency models extended image saliency networks with mechanisms for motion and memory. Bak et al. [70] used two-stream networks, one stream on the frames and one on optical flow, and compared ways of fusing them. Lai et al. [71] coupled appearance and motion streams more tightly through dense residual cross connections and added a convolutional gated recurrent unit for long-term dynamics. Gorji and Clark [72] took the opposite route and augmented a strong static saliency model with recurrent "attentional push" pathways, which model how the photographer and the actors in a scene direct the viewer's attention.

Recurrence on top of per-frame convolutional features became the dominant design. DeepVS [54] combined an object-to-motion network with a saliency-structured convolutional LSTM, motivated by the observation, made on its own large eye-tracking dataset, that moving objects attract gaze. ACLNet [8], [9], introduced together with DHF1K, added a supervised attention module that encodes static saliency to a CNN-LSTM, so that the recurrent part can concentrate on temporal change; it was trained with static images from SALICON alongside the videos. Linardos et al. [73] showed that a simple exponential moving average of features can perform on a par with a convolutional LSTM, which suggests that much of the temporal modelling in these networks amounts to smoothing. UNISAL [74] pursued a lightweight encoder–RNN–decoder design trained jointly on image and video saliency data, with domain-adaptive priors, fusion and smoothing to cope with the differences between datasets, and reached strong results at a fraction of the size of its competitors.

These models share a limitation: their spatial features come from networks pretrained on images, and their temporal modelling is confined to a recurrent state on top of those features.

Their reliance on static saliency data, which helped them markedly [9], is a sign that the video data alone were not sufficient to train them.

3.2. Three-Dimensional Convolutional Encoder–Decoders

A second generation replaced per-frame features by spatio-temporal features from three-dimensional convolutional networks pretrained for action recognition, typically S3D [36] trained on Kinetics [35]. TASED-Net [75] encodes a clip of consecutive frames and decodes the features spatially while aggregating the temporal dimension, producing one map per clip; maps for a whole video are obtained by sliding the window, under the assumption that a limited number of past frames suffices to predict the current one. HD2S [76] supervises intermediate "conspicuity" maps at several levels of abstraction and adds domain adaptation through gradient reversal and dataset-specific priors. ViNet [77] decodes features from several levels of an action-recognition encoder with trilinear upsampling and three-dimensional convolutions; it is causal, runs in real time and, despite using no audio, outperformed audio-visual models on six audio-visual datasets.

Later work in this family focused on efficiency and on the choice of encoder. TinyHD [78] combines several simple heterogeneous decoders with hierarchical map distillation from a larger teacher, and predicts several output frames per forward pass to reduce cost. MSFF-Net [79] fuses multi-scale spatio-temporal features through a bi-directional pyramid with attention-guided fusion. Most recently, ViNet-S and ViNet-A [80] paired a lightweight decoder with either an S3D encoder or a SlowFast encoder pretrained for spatio-temporal action localisation; their ensemble, ViNet-E, reported the best results among the compared methods on UCF Sports and Hollywood-2. The strength of the action-localisation encoder on these two datasets, whose gaze was recorded during action recognition [56], illustrates how much the pre-training of the encoder matters.

Three-dimensional convolutions capture motion well, but their receptive field in time is limited by the kernel sizes, and their encoders were trained for classification on a fixed set of action categories.

3.3. Transformer-Based Models

Self-attention relaxes the locality of convolutions. STSANet [81] inserts spatio-temporal self-attention modules at several levels of a three-dimensional convolutional backbone to capture long-range relations between time steps, and fuses the levels with an attentional multi-scale module. VSFT [82] was the first purely transformer-based model in the field and extended the task to forecasting the saliency of future frames. TMFI-Net [83] builds on multi-level features of a video transformer, enriches them top-down with high-level semantics in a semantic-guided encoder, and decodes them hierarchically with attention along the channel, temporal and

spatial dimensions. THTD-Net [84] argues that the temporal dimension of transformer features should be kept high throughout decoding, and reaches results comparable to multi-branch decoders with a single-branch one. UniST [85] uses one saliency-aware transformer for both video saliency prediction and video salient object detection, and TM2SP [86] combines a Video Swin encoder [42] with a multi-level spatio-temporal feature pyramid guided by a pretrained image saliency encoder.

These models raised the published DHF1K test correlation from about 0.52 to about 0.55 [10]. They remain, however, trained end to end, or with an encoder pretrained for supervised action recognition, on the few hundred videos of a saliency dataset, and they are large: TMFI-Net has about 61 million parameters and STSANet about 169 million [80].

3.4. Video Foundation Models for Saliency

Foundation models trained with self-supervision on large video collections offer a different starting point. SalFoM [10] was the first video saliency model built on such an encoder. It fine-tunes the UMT-L encoder [48], a ViT-L trained by masked video modelling guided by an image–text teacher, together with a heterogeneous decoder that includes a locality-aware spatio-temporal transformer. Given 16 frames at 224×224 pixels, it predicts the saliency map of the last frame, and maps for a whole video are produced with a sliding window. SalFoM reported the best DHF1K test results among the methods it compared with (CC 0.569, NSS 3.353), but it was weaker than earlier models on UCF Sports, which its authors attribute to the task-driven gaze of that dataset and to overfitting on its small training set [10].

The same group has since combined foundation encoders with explicit long- and short-term temporal modelling [87] and distilled them into smaller models to reduce their cost [88]. A different direction couples a multimodal large language model with a saliency decoder to produce saliency maps together with textual explanations, trained on a new dataset of gaze annotated with reasons [89].

Two properties of these approaches are relevant here. First, the encoder is adapted to the saliency data, either by full fine-tuning or by distillation; whether a frozen foundation encoder is sufficient, and how it compares with an adapted one, was not examined in the studies whose full texts could be consulted. Second, the sliding-window design runs the large encoder once for every output frame, which makes inference expensive.

3.5. Audio-Visual Saliency Models

Because most videos have sound, and sound influences where people look [2], [25], a separate line of work fuses the soundtrack with the frames. STAViS [63] combined a spatio-temporal visual saliency network with auditory features, learned to localise sound sources, and fused the two saliencies; it also introduced the protocol, with three folds per

dataset, under which six audio-visual datasets are now evaluated. CASP-Net [90] addresses the temporal inconsistency between the two modalities with a consistency-aware predictive coding module, and DiffSal [64] formulates prediction as conditional generation with a diffusion model conditioned on audio-visual features. Models based on implicit neural representations map space–time coordinates to saliency values and fuse the modalities parametrically [91], [92], and the relevance-guided fusion of Yu et al. [93] adjusts how much audio is retained according to its semantic relevance to the visual content.

Whether these models actually use the audio is disputed. Jain et al. [77] found that an audio-augmented variant of ViNet became agnostic to its audio input after training and observed the same behaviour in an earlier audio-visual model. Agrawal et al. [94] examined a range of audio modules and fusion techniques on ten datasets and found that none improved on visual-only models. On several audio-visual datasets, visual-only models such as ViNet-E report results that equal or exceed those of audio-visual models [80].

3.6. Prediction Error as an Attention Signal

The idea that attention is drawn to what violates expectation has a long history. Itti and Baldi formalised "surprise" as the divergence between an observer's prior and posterior beliefs, detected surprising events in video with low-level feature models [26], and showed that this measure predicted human gaze shifts better than the other metrics they tested [13]. Predictive-coding accounts of the visual cortex propose that neural responses signal the error of predictions made from context [27], and deep networks built on this principle have been trained to predict future video frames [95].

Modern self-supervised video models contain an explicit predictor. In the joint-embedding predictive architectures I-JEPA [11] and V-JEPA [12], a predictor network learns to predict the representations of masked regions from the visible ones. Such a predictor offers, in principle, a learned and high-level analogue of surprise: its error indicates which parts of a scene the model could not anticipate. In the literature reviewed for this thesis, the error of such a learned latent predictor has not been evaluated against human fixations. CASP-Net uses predictive coding in a different sense, to align audio and visual features [90].

3.7. Summary and Research Gap

Table 3.1 summarises the most relevant works. Three observations emerge from the review.

First, progress in video saliency prediction has tracked progress in the encoder: from image networks with recurrence, through action-recognition three-dimensional networks, to video transformers and, most recently, video foundation models. The decoders, by contrast, have converged on variations of multi-scale fusion. Yet the encoders used for saliency have all been trained with supervision or with objectives that reconstruct or align tokens, and they

Table 3.1: Summary of the closest related works

Work	Approach	Evaluation	Limitation relevant to this thesis
ACLNet [9]	CNN-LSTM with static-saliency attention	DHF1K, Hollywood-2, UCF	image features; needs static data
UNISAL [74]	lightweight encoder–RNN–decoder, image and video jointly	DHF1K, Hollywood-2, UCF, SALICON	recurrent temporal model on image features
TASED-Net [75]	S3D encoder, temporally aggregating decoder	DHF1K, Hollywood-2, UCF	one map per clip; action-recognition encoder
ViNet-S/A/E [80]	S3D or action-localisation encoder, light decoder	DHF1K val, Hollywood-2, UCF, six AV sets	supervised encoders; one map per clip
TinyHD [78]	heterogeneous decoders, distillation, multi-output	DHF1K, Hollywood-2, UCF	efficiency focus; CNN encoder
STSANet [81]	self-attention on a 3D CNN	DHF1K, Hollywood-2, UCF, DIEM	large (about 169 M parameters)
TMFI-Net [83]	video transformer, multi-scale integration	four video datasets	trained on few videos; large
SalFoM [10]	fine-tuned UMT-L, heterogeneous decoder	DHF1K test, Hollywood-2, UCF	full fine-tuning; one encoder pass per output frame
STAViS [63]	visual saliency with sound localisation and fusion	six AV datasets	audio benefit later disputed [94]
DiffSal [64]	audio-visual conditional diffusion	six AV datasets	iterative generation; audio-dependent design
Itti and Baldi [13]	Bayesian surprise of low-level features	gaze in video	hand-crafted features, no learned predictor

have been adapted to the saliency data. Encoders trained with a feature-prediction objective, such as V-JEPA [12], whose authors report that its features perform well without adapting the model's parameters, have not been evaluated for this task in the peer-reviewed studies whose full texts could be consulted.

Second, the evaluation practice of the field makes small differences hard to interpret. DHF1K results are reported on the validation or the test set, published numbers for the same model differ between papers by up to 0.03 CC [80], [85], and comparisons are almost always made against numbers copied from other papers rather than against re-scored predictions with an uncertainty estimate.

Third, the idea that motivated this work, that a learned video model's prediction error marks where people look, has support in classical models of surprise [13] but had not been tested with the predictor of a modern self-supervised video model.

This thesis addresses these gaps. It evaluates a frozen V-JEPA 2 encoder with a light decoder that predicts many frames per pass, compares it with SalFoM, the closest foundation-model approach, under one validated evaluator with paired statistics, tests the encoder's own predictor as a source of attention cues, and examines which design choices still matter once the encoder is fixed. It resembles SalFoM in building on a video foundation encoder and shares its multi-scale decoder design with much of the field; it differs in keeping the encoder frozen, in the objective with which that encoder was trained, in the multi-frame output, and in the pre-registered experimental protocol.

CHAPTER FOUR

METHODOLOGY

This chapter describes how the research questions of Section 1.4 were investigated. It first sets out the research design, which fixes every decision rule before the experiment that it governs (Section 4.1), and formalises the prediction task (Section 4.2). It then describes the eye-tracking data and their preparation (Section 4.3), the proposed model, SURGE-D, as it is implemented (Section 4.4), the components that were implemented and tested but not retained (Section 4.5), the training and inference procedures (Sections 4.6 and 4.7), and the evaluation metrics and statistical analysis (Section 4.8). The chapter closes with the ethical considerations of the work (Section 4.9). Every equation in this chapter corresponds to code in the accompanying repository; where the implementation departs from the original research proposal, the departure is stated.

4.1. Research Design

The study follows a design-science approach in which an artefact, a video saliency model, is built and then evaluated through controlled experiments. Two features of the design deserve emphasis, because they determine how the results in Chapter 5 may be read.

Pre-registered decision rules. The original proposal contained several components whose value was uncertain: the choice of video encoder, a "surprise" signal derived from the encoder's predictor, parameter-efficient adaptation of the encoder, static-image co-training, and others. Rather than include all of them and report the final configuration, each component was treated as a hypothesis with an explicit test. Before each experiment was run, its decision rule (the quantity to be measured, the data on which it is measured, and the threshold that decides the outcome) was written into a dated research log, and the rule was applied unchanged afterwards. Table 4.1 lists these rules. This procedure guards against two familiar sources of optimism in deep-learning research: choosing among variants after seeing test results, and adjusting thresholds after seeing which side of them a result falls on.

Separation of selection and testing. All model selection (the best training epoch, the output blur, every gate in Table 4.1) was performed on internal holdout sets carved out of the training data. Each test set was scored once per reported model, after all of that model's settings had been fixed. For DHF1K, whose official test annotations are withheld by the dataset authors, the 100-video validation set serves as the test set and was never used for selection.

Table 4.1: Decision rules, each written before the experiment that it governs, except the last (see text). Holdout CC is measured on the internal 60-video DHF1K holdout unless stated otherwise; outcomes are reported in Chapter 5

Decision	Rule
Evaluator validity	Re-scoring the same maps with an independent evaluator must agree within 0.002 (CC, SIM) and 0.01 (NSS, KLD)
Backbone (gate G1)	Highest holdout CC wins; the proposal's choice, V-JEPA 2.1, is kept if within 0.005 CC of the winner
Surprise (gate G2)	Enabled only if a surprise map beats both token atypicality and frame difference on NSS and has positive information gain over the centre prior
LoRA adaptation	Adopted if the three-seed mean holdout gain is at least +0.003 CC and LoRA wins in at least two of three seeds
Single-seed ablations	A difference is reported only if $\|\Delta\text{CC}\| \geq 0.003$ on the holdout
Claim over a baseline	The 95 % bootstrap interval of ΔCC over test videos must lie above zero
ViT-g backbone	Adopted if its holdout CC is at least 0.5833 (the ViT-L three-seed mean plus 0.005)
Training on 600 videos	Replaces the 540-video models if the ensemble's ΔCC interval is above zero and it is worse on at most two of six metrics
32-frame clips	Pursued if the holdout CC is at least 0.5904 (the 16-frame three-seed mean plus 0.005)
Audio fusion	"Helps" on a dataset only if the fold-mean ΔCC is at least +0.010 and matched audio beats rotated audio by at least 0.005
Published results*	Comparisons with numbers from other evaluation code treat margins below 0.01 CC as ties

The rule for published results (marked *) differs from the others in one respect. The rule written before the first audio-visual experiment required only a higher fold-mean CC than the published value, and noted that cross-paper numbers disagree by a few hundredths. The tie margin of 0.01 was introduced when the first audio-visual results were interpreted and was then applied unchanged to every later comparison with published numbers.

Unit of analysis. Saliency metrics are computed per frame and averaged within each video; the video is the unit of analysis, and uncertainty is estimated by resampling videos (Section 4.8.2). The margins of the early rules (0.003 for single-seed ablations and 0.005 for gate G1) were set by judgement before any seed variation had been measured. The seed-to-seed spread later observed for the main model, about 0.003 CC, informed the margins of the later gates for ViT-g and for 32-frame clips.

4.2. Problem Formulation

Let a video be a sequence of frames $I_1, \ldots, I_N$ with $I_n \in \mathbb{R}^{H \times W \times 3}$. For each frame, an eye-tracking dataset provides a binary fixation map $F_n \in \{0, 1\}^{H \times W}$, marking the pixels fixated by at least one observer, and a continuous fixation density Y_n, obtained by blurring the fixations with a Gaussian kernel. Video saliency prediction is the task of estimating, for every frame, a

probability distribution $\hat{S}_n$ over pixel locations,

$$\hat{S}_n(p) \geq 0, \qquad \sum_p \hat{S}_n(p) = 1, \tag{4.1}$$

that approximates where observers look in frame n. Treating saliency as a distribution, rather than as an unnormalised score map, is the probabilistic view advocated by Kümmerer et al. [68], [69]; it makes the training objective and the distribution-based metrics well defined.

The model sees a temporal window rather than a single frame. A clip of $T = 16$ frames is sampled with a stride of $s = 2$, so that the sampled indices are $n_k = n_0 + ks$ for $k = 0, \ldots, T-1$; at the 30 frames-per-second rate of DHF1K the clip spans 31 original frames, or about one second. The stride is defined in frames, so on datasets with other frame rates the clip covers a correspondingly different duration. Indices that fall outside the video are clamped to its first or last frame. The model predicts maps for all T sampled frames but only the central $T/2$ positions,

$$O = \{T/4, \ldots, 3T/4 - 1\} = \{4, \ldots, 11\}, \tag{4.2}$$

are used for training and inference. Each prediction therefore has at least eight original frames (about 0.27 s) of context on either side. The formulation is thus *non-causal*: a frame's map depends on frames that follow it. This suits offline applications such as video compression or summarisation but not streaming use, and it differs from methods that predict the last frame of a clip from past frames only, such as SalFoM [10]. The difference is returned to when results are compared in Chapter 5.

4.3. Datasets

4.3.1. Data Collection

No new eye-tracking data were collected. The study uses ten public datasets, summarised in Table 4.2. All were read from their original distributions; the splits and holdout lists used here are provided with the code.

DHF1K [8], [9] is the primary benchmark. Its 600 training videos were split once, with a fixed random seed, into 540 training videos and a 60-video holdout used for all model selection. The official validation set (videos 601–700, 60,273 frames) is the test set of this thesis. Predictions for the 300 official test videos were generated with the final model and packaged in the format required by the dataset authors; their scores were not available when this thesis was written and are not reported.

UCF Sports [57] and Hollywood-2 [55] are action-recognition datasets for which Mathe and Sminchisescu [56] recorded the gaze of observers performing action recognition. For UCF Sports the standard 103/47 split was used, with ten training videos held out. The Mathe and Sminchisescu ground truth for Hollywood-2 could not be obtained during the project;

Table 4.2: Eye-tracking datasets used in this thesis. "Holdout" is the internal selection set drawn from the training part; the test part is scored once per model

Dataset	Content and viewing condition	Split used (train / holdout / test)
DHF1K [8], [9]	1,000 YouTube videos, 640 × 360 at 30 fps, free viewing, 17 observers; official split 600 / 100 / 300 (test labels withheld)	540 / 60 / 100 (official validation set as test)
UCF Sports [56], [57]	150 sports videos; gaze recorded during action recognition; standard split 103 / 47	93 / 10 / 47
Hollywood-2 [55] with the gaze of Vig et al. [58]	1,707 movie clips (823 / 884); four observers who watched every clip plus a pooled fifth observer, recorded during action recognition	741 / 82 / 884
DIEM [1]	84 videos of many genres; audio present	one standard split, 64 training and 17 test videos
Coutrot1 [2], [25]	60 clips in four visual categories	three folds each, every video tested once [63]; holdout = 10 % of each training fold
Coutrot2 [2], [25]	15 clips of four people in a meeting	
ETMD [59], [60]	12 clips from six Hollywood movies	
AVAD [61]	45 short audio-visual clips	
SumMe [60], [62]	25 user-made videos	
SALICON [28]	10,000 static training images	co-training ablation only

instead, the independently recorded and openly licensed gaze data of Vig, Dorr and Cox [58] were used[1]. These data come from fewer observers and cover whole clips rather than the shot-segmented clips used in the saliency literature, so results on them are reported under the separate name "Hollywood-2 (Vig gaze)" and are not compared with published Hollywood-2 numbers.

The six audio-visual datasets (DIEM [1], Coutrot1 and Coutrot2 [2], [25], ETMD [59], AVAD [61] and SumMe [62]) were used in the preprocessed form and with the fold lists released by Tsiami et al. [63], whose protocol has become the standard for these data: DIEM has one fixed split, and each of the other five datasets has three folds in which every video appears in exactly one test set. The gaze data for ETMD and SumMe were recorded by Tsiami et al. [60]. SURGE-D is a visual model; the soundtracks were used only in the audio experiment of Section 4.5.3. SALICON [28], a dataset of static images with mouse-tracking saliency, was used only in the static co-training ablation.

[1]Figshare item 93951, released under the CC BY licence; every file was checked against its published checksum.

4.3.2. Preprocessing

Frames. Frames were read as distributed (DHF1K, UCF Sports and the STAViS releases) or decoded once with OpenCV (Hollywood-2, 487,556 frames). For the model, every frame is resized to 256×256 pixels with area interpolation, which changes the aspect ratio of 16:9 video, and normalised with the ImageNet channel means and standard deviations [96]. Squashing was chosen over cropping because cropping removes image borders and alters the centre-bias statistics of the fixation maps; its effect on accuracy was not ablated.

Training targets. For each target frame, the ground-truth density is resized with area interpolation to the 64×64 output grid and normalised to sum to one, and the fixated pixels of the binary fixation map are counted on the same grid. The resulting pair (Y_t, F_t) is used by the loss (Section 4.6). Evaluation, in contrast, always takes place at the native resolution of the ground truth (Section 4.8).

Frames without fixations. In the STAViS releases and in the Hollywood-2 (Vig gaze) maps, some frames have no recorded fixation, for example during blinks. These frames are excluded from scoring. DHF1K and UCF Sports contain no such frames.

Hollywood-2 (Vig gaze) maps. The Vig gaze recordings provide raw 1000 Hz samples with confidence values and saccade annotations. A frame f of a clip with frame rate r was taken to cover the interval $[(f-1)/r,\ f/r)$. For each observer and frame, the samples with confidence 1.0 that do not fall inside an annotated saccade were collected. If any remained, the observer contributed one fixation point, the median of those samples, rounded and clipped to the frame. The binary fixation map thus has at most five points per frame. The density map is this binary map blurred with a Gaussian of $\sigma = 20$ pixels, following the DHF1K recipe [8]. Of the 487,556 frames, 99.8 % received at least one point, with 4.53 points per frame on average. For model selection, 82 of the 823 training clips were held out with a fixed seed.

Audio features. For the audio experiment only, each soundtrack of the STAViS release was mixed to mono and resampled to 16 kHz, and the frozen Audio Spectrogram Transformer [97] fine-tuned on AudioSet [98] was applied to consecutive 10.24 s chunks[2]. Averaging the patch tokens over the twelve frequency rows of the patch grid gives one 768-dimensional vector per time column, that is, roughly one every 0.1 s. The feature of video frame k (centred at time $(k - \frac{1}{2})/r$) is the mean of the columns whose centres lie within ±0.5 s of that time. Videos without a soundtrack receive zero features.

[2]Checkpoint `MIT/ast-finetuned-audioset-10-10-0.4593`, BSD-3-Clause licence.

4.4. Proposed Architecture

SURGE-D consists of a frozen self-supervised video encoder and a light trainable decoder (Figure 4.1). The decoder reads four intermediate layers of the encoder, rebuilds a multi-scale feature pyramid, decodes it top-down with factorised spatio-temporal convolutions, and adds a learned centre prior before normalising each output frame into a probability distribution. Only the decoder and the prior are trained: 6.99 million parameters with the ViT-L encoder and 7.39 million with ViT-g. The name derives from the original design of the project, a *SURprise-Gated Encoder*–decoder in which the predictor's error was to modulate the decoder; the suffix D marks the reduced design that remained after the tests of Section 4.5, in which the surprise branch did not pass its gate.

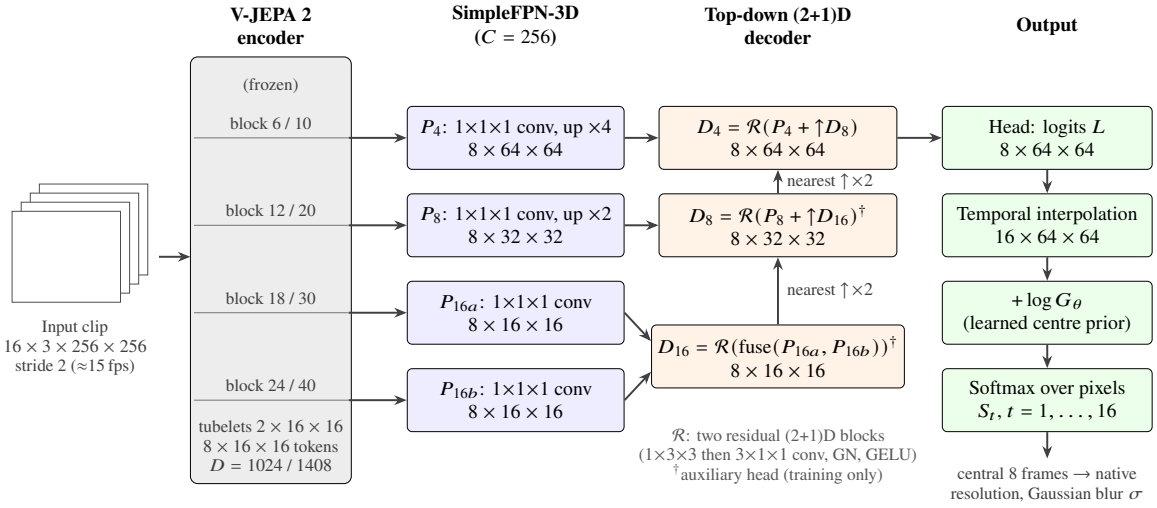


Figure 4.1: Architecture of SURGE-D. A clip of 16 frames passes through the frozen V-JEPA 2 encoder; four intermediate layers (blocks 6/12/18/24 of ViT-L or 10/20/30/40 of ViT-g) feed a three-dimensional simple feature pyramid, which a top-down (2+1)D decoder turns into logits at 64 × 64 resolution for eight tubelet steps. After temporal interpolation to 16 frames and the addition of a learned centre prior, a softmax over pixels yields one saliency distribution per frame; the central eight frames are kept. Tensor sizes are given as time × height × width

4.4.1. Frozen Video Encoder

The encoder is a Vision Transformer [39] trained with the video joint-embedding predictive architecture (V-JEPA) objective of Bardes et al. [12]. In this objective, which extends the image model I-JEPA [11] to video, parts of a clip are masked and a predictor network is trained to predict the *representations* of the masked regions, produced by an exponential-moving-average target encoder, from the representations of the visible ones. Because the loss is computed in representation space rather than pixel space, the encoder is not required to model unpredictable pixel detail, which Bardes et al. argue leads to features that transfer well with a frozen backbone [12].

The checkpoints used are the publicly released V-JEPA 2 models[3], which follow this objective: as Mur-Labadia et al. describe it, an L_1 loss between the predictor's outputs and the target encoder's representations is applied to the masked tokens only [14]. The same study reports that this choice leaves the visible tokens without a direct training signal and that frozen V-JEPA 2 features consequently carry little local structure, reaching only 22.2 mIoU in linear-probe semantic segmentation on ADE20K. Whether such features suit a dense task such as saliency prediction was therefore an open question, which gate G1 addresses (Section 5.3). Two sizes were used. ViT-L/16 has 24 transformer blocks of width $D = 1024$ and 303.9 million encoder parameters; ViT-g/16 has 40 blocks of width $D = 1408$ and 1,012.2 million parameters (both counts measured on the loaded checkpoints, excluding the predictor). Both embed the video in *tubelets* of $2 \times 16 \times 16$ pixels, so a $16 \times 256 \times 256$ clip becomes $8 \times 16 \times 16 = 2{,}048$ tokens.

Let $h^{(\ell)} \in \mathbb{R}^{2048 \times D}$ denote the output of block ℓ. Four blocks at equal fractions of the depth are tapped, $\ell \in \{6, 12, 18, 24\}$ for ViT-L and $\ell \in \{10, 20, 30, 40\}$ for ViT-g, and each tap is reshaped into a feature volume of size $D \times 8 \times 16 \times 16$. Intermediate layers are used because dense prediction benefits from features at several levels of abstraction [43], [44]. The encoder is frozen: its weights are not updated and no gradient is propagated into it. Freezing was the outcome of an experiment rather than an assumption; the alternative, low-rank adaptation of the encoder, is described in Section 4.5.2 and evaluated in Chapter 5.

4.4.2. Three-Dimensional Simple Feature Pyramid

A plain Vision Transformer produces features at a single resolution. Li et al. [45] showed that a feature pyramid for dense prediction can be built from such a backbone by simple up- and down-sampling of its features. SURGE-D uses a three-dimensional variant built from the four taps rather than from the last layer only. With $\phi(\cdot)$ denoting group normalisation [99] followed by a GELU non-linearity, W_ℓ a $1 \times 1 \times 1$ convolution from D to $C = 256$ channels, and $U(\cdot)$ a spatial $\times 2$ transposed convolution (kernel and stride $1 \times 2 \times 2$) followed by ϕ, the pyramid levels are

$$\begin{aligned} P_4 &= U\Big(U\Big(\phi(W_1 h^{(\ell_1)})\Big)\Big) \in \mathbb{R}^{C \times 8 \times 64 \times 64}, & P_8 &= U\Big(\phi(W_2 h^{(\ell_2)})\Big) \in \mathbb{R}^{C \times 8 \times 32 \times 32}, \\ P_{16a} &= \phi(W_3 h^{(\ell_3)}) \in \mathbb{R}^{C \times 8 \times 16 \times 16}, & P_{16b} &= \phi(W_4 h^{(\ell_4)}) \in \mathbb{R}^{C \times 8 \times 16 \times 16}. \end{aligned} \tag{4.3}$$

The shallowest tap feeds the finest level, as in standard feature pyramids, and the two deepest taps both remain at the token resolution of $1/16$. The temporal resolution of eight tubelet steps is preserved throughout.

[3] `facebook/vjepa2-vitl-fpc64-256` and `facebook/vjepa2-vitg-fpc64-256`, released under the MIT licence through the Hugging Face hub and loaded with the Transformers library [16]. The technical report that introduced them had not appeared in a peer-reviewed venue when this thesis was written and is therefore not cited; their properties are taken from the checkpoints themselves and from the peer-reviewed follow-up study of Mur-Labadia et al. [14].

4.4.3. Top-Down (2+1)D Decoder

The decoder refines the pyramid from coarse to fine. Its building block is a residual block that factorises a three-dimensional convolution into a spatial and a temporal part, in the manner of R(2+1)D [37],

$$\mathcal{B}(x) = \mathrm{GELU}\Big(x + \mathrm{GN}\big(\mathrm{Conv}_{3\times1\times1}\big(\mathrm{GELU}(\mathrm{GN}(\mathrm{Conv}_{1\times3\times3}(x)))\big)\big)\Big), \tag{4.4}$$

where the subscripts give the kernel size as time × height × width and GN is group normalisation. Let $\mathcal{R}$ denote two such blocks in sequence and $\uparrow$ nearest-neighbour spatial upsampling by a factor of two. The decoder computes

$$\begin{aligned} D_{16} &= \mathcal{R}\big(\phi(\mathrm{Conv}_{1\times1\times1}[P_{16a}; P_{16b}])\big), \\ D_8 &= \mathcal{R}\big(P_8 + \uparrow D_{16}\big), \\ D_4 &= \mathcal{R}\big(P_4 + \uparrow D_8\big), \end{aligned} \tag{4.5}$$

where $[\cdot;\cdot]$ concatenates channels. A head with a $1 \times 3 \times 3$ convolution to 128 channels, a GELU and a $1 \times 1 \times 1$ convolution to one channel turns D_4 into logits $L \in \mathbb{R}^{8\times64\times64}$. During training, two auxiliary heads ($1 \times 1 \times 1$ convolutions) produce additional logits from D_{16} and D_8; they are upsampled to 64×64 and supervised with the same targets, a form of deep supervision [100]. The auxiliary heads are not used at inference.

4.4.4. Temporal Upsampling, Centre Prior and Output

The logits have one time step per tubelet. They are interpolated trilinearly to the $T = 16$ sampled frames, giving L_t for $t = 1, \ldots, T$.

Human fixations in videos are biased towards the centre of the screen, partly because of how scenes are framed and partly because of viewing strategy [5], [24]. SURGE-D models this bias with a learned, input-independent prior. With pixel positions $p \in [0, 1]^2$ in normalised coordinates, the prior is a mixture of $K = 8$ Gaussians with diagonal covariance,

$$\log G_\theta(p) = \alpha \left[\log \sum_{k=1}^{K} \pi_k \, \mathcal{N}\Big(p; \mu_k, \mathrm{diag}(\sigma_k^2)\Big) - \log \sum_{p'} \sum_{k=1}^{K} \pi_k \, \mathcal{N}\Big(p'; \mu_k, \mathrm{diag}(\sigma_k^2)\Big) \right], \tag{4.6}$$

where the mixture weights $\pi = \mathrm{softmax}(w)$, the means μ_k, the scales σ_k and the strength α are learned (41 parameters in total). The means are initialised on a grid in the central $[0.35, 0.65]^2$ region, the scales at 0.25 and the strength at one. The prediction for frame t is

$$S_t(p) = \frac{\exp\big(L_t(p) + \log G_\theta(p)\big)}{\sum_{p'} \exp\big(L_t(p') + \log G_\theta(p')\big)}, \tag{4.7}$$

a softmax over the 64×64 positions, so that every output frame is a probability distribution as

required by Equation (4.1). The prior is also added to the auxiliary logits during training.

The design of the decoder follows from three considerations. First, the encoder carries most of the representational capacity, so the decoder can be small; its 7 million parameters are about 2 % of the ViT-L encoder. Second, the (2+1)D factorisation adds temporal mixing at every scale at a lower cost than full three-dimensional kernels [37]. Third, predicting maps for many frames in one forward pass amortises the cost of the encoder: a single pass over a 16-frame clip yields usable maps for 16 original frames (Section 4.7). The measured cost is reported in Section 5.5.4.

4.5. Components Evaluated but Not Retained

Four further components were implemented, tested under the rules of Table 4.1, and not retained. They are described here because their negative results are part of the contribution of the thesis.

4.5.1. Predictive Surprise from the V-JEPA 2 Predictor

The original proposal was built around the hypothesis that the prediction error of a self-supervised video model indicates where something unexpected happens, and therefore where people look, in the spirit of the Bayesian surprise account of attention [13]. V-JEPA 2 provides the means to test this, because its released checkpoint includes the predictor used during pre-training.

For a window of 16 sampled frames, the first eight frames (four tubelets) are the context and the last eight the target. The context is encoded on its own, so that no information from the target frames leaks into it, and the predictor maps the context tokens to predicted target tokens $\hat{z}_p$. The target tokens z_p are taken from an encoding of the full window. With LN a parameter-free layer normalisation over the channel dimension, the *raw surprise* of target token p is

$$e(p) = \frac{1}{D} \sum_{c=1}^{D} \left| \mathrm{LN}(\hat{z}_p)_c - \mathrm{LN}(z_p)_c \right|. \tag{4.8}$$

A *contextual* variant subtracts the error obtained when the context is replaced by the context of a different video, $e(p) - e_{\mathrm{wrong}}(p)$, so that only the part of the error that depends on the true past remains. Two reference maps tested whether surprise adds anything beyond simpler cues: *token atypicality*, the mean absolute deviation of each target token from the mean token of its frame, and *frame difference*, the absolute grey-level difference between consecutive frames pooled to the 16×16 token grid. All maps were upsampled to 640×360, blurred with a Gaussian of $\sigma = 16$ pixels and scored on the first frame of each target tubelet of the 60 holdout videos, at about 4 and 15 frames per second. A decoder module that injects the surprise map through zero-initialised feature-wise modulation was implemented but, following the outcome of the gate, never trained.

4.5.2. Low-Rank Adaptation of the Encoder

Low-rank adaptation (LoRA) [51] fine-tunes a frozen linear layer $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ by adding a trainable low-rank update,

$$W'x = Wx + \frac{\alpha}{r} BAx, \qquad A \in \mathbb{R}^{r \times d_{\text{in}}},\ B \in \mathbb{R}^{d_{\text{out}} \times r}, \tag{4.9}$$

with B initialised to zero so that training starts from the frozen model. LoRA with $r = \alpha = 16$ was applied to the query, key, value and output projections of the last twelve blocks of ViT-L (48 layers, 1.57 million parameters). Each LoRA model was initialised from the frozen model of the same seed and trained for three epochs with learning rates of 10^{-4} for the LoRA weights and 10^{-5} for the decoder.

4.5.3. Audio Fusion

To test whether the soundtrack helps on the audio-visual datasets, an audio fusion module was added to three pyramid levels (P_8, P_{16a}, P_{16b}). For each tubelet, the audio vector a (the mean of the features of its two frames) is embedded as $e = \text{GELU}(W_p\,\text{LN}(a)) \in \mathbb{R}^{256}$. A pyramid level F is then modulated as

$$\tilde{F} = F \odot (1 + \gamma) + \beta + W_r\big(F \odot s\big), \qquad s(x) = \text{sigmoid}\left(\frac{\langle W_q F(x),\ W_k e\rangle}{\sqrt{256}}\right), \tag{4.10}$$

where $(\gamma, \beta) = W_f e$ is a feature-wise linear modulation (FiLM) [52], $s(x)$ is an audio-visual correspondence map at each position x, and W_r is a $1 \times 1 \times 1$ convolution. Because W_f and W_r are initialised to zero, the module is exactly the visual model at the start of training. As a control, the trained model was also scored with the audio of each test video replaced by that of another test video ("rotated audio"); a model that uses sound should do better with the matched soundtrack.

4.5.4. Static-Image Co-Training

Training on static images helped early video saliency models [8]. To test whether it still helps on top of a video foundation encoder, the 10,000 SALICON training images [28] were added to each training epoch, each image repeated over the 16 input frames, with the same loss.

4.6. Training Procedure

Loss. The loss combines three standard saliency objectives: the Kullback–Leibler divergence of the prediction from the ground-truth density, the linear correlation coefficient, and the normalised scanpath saliency [17]. For a predicted distribution S, a target density Y and

fixation counts F on the 64×64 grid,

$$\mathrm{KL}(Y \parallel S) = \sum_p Y(p)\big[\log(Y(p)+\epsilon) - \log S(p)\big], \tag{4.11}$$

$$\mathrm{CC}(S,Y) = \frac{\sum_p (S(p)-\bar{S})(Y(p)-\bar{Y})}{\big\|S-\bar{S}\big\|_2 \big\|Y-\bar{Y}\big\|_2 + \epsilon}, \tag{4.12}$$

$$\mathrm{NSS}(S,F) = \frac{1}{\sum_p F(p)} \sum_p F(p)\,\frac{S(p)-\bar{S}}{\mathrm{std}(S)}, \tag{4.13}$$

with $\epsilon = 10^{-8}$. The training loss over the central frames O of a clip is

$$\begin{aligned}\mathcal{L} = \frac{1}{|O|}\sum_{t\in O}\Big[\mathrm{KL}(Y_t\|S_t) + \big(1-\mathrm{CC}(S_t,Y_t)\big) - 0.1\,\mathrm{NSS}(S_t,F_t)\Big] \\ + 0.25 \sum_{a\in\{8,16\}} \frac{1}{|O|}\sum_{t\in O}\Big[\mathrm{KL}(Y_t\|S_t^a) + \big(1-\mathrm{CC}(S_t^a,Y_t)\big)\Big],\end{aligned} \tag{4.14}$$

where S_t^a are the predictions of the auxiliary heads and the NSS term is averaged over frames with at least one fixation. The KL and CC terms follow the loss of SalFoM [10]; the small NSS term additionally rewards high predicted values at the fixated locations themselves, which the two distribution-based terms do not address directly. The loss is averaged over the clips of a batch.

Optimisation. The decoder was trained with AdamW [101] at a learning rate of 3×10^{-4} and weight decay 0.01; the prior used a learning rate of 10^{-3} without weight decay. The learning rate rose linearly during the first quarter epoch and then followed a cosine decay to zero [102]. Gradients were clipped to a norm of 1.0, and the forward pass ran in bfloat16 mixed precision. Each epoch drew eight random clips from every training video, and the batch size was eight. Clip start positions were drawn uniformly, allowing up to a quarter of the clip to extend beyond the start or end of the video, where frames are repeated. The only data augmentation was a horizontal flip of frames and targets with probability 0.5; the colour jitter mentioned in the original proposal was not implemented.

Schedule and model selection. The proposal foresaw 20 epochs. In the backbone comparison (Section 5.3), however, the holdout CC of every backbone peaked after the second or third epoch and then fell while the training CC kept rising. The schedule for all later runs was therefore fixed, before any of them started, at three epochs, with the checkpoint chosen by holdout CC after each epoch (computed on every fifth frame).

Fine-tuning on other datasets. For UCF Sports, the six audio-visual datasets and Hollywood-2 (Vig gaze), each model was initialised from the DHF1K model and trained for three further epochs with a learning rate of 10^{-4} (3×10^{-4} for the prior). The number of clips per training

video was set so that an epoch contained about 1,500 clips (16 clips per video for UCF Sports). Each fold of the audio-visual datasets was trained and selected on its own holdout and its test videos were scored once.

Seeds and ensembles. The main models were trained with three seeds (0, 1 and 2). A three-seed ensemble averages the members' probability maps before post-processing. On the fine-tuning datasets, ensemble member k is fine-tuned from the DHF1K model of seed k. Table 4.3 summarises the settings.

Table 4.3: Training and inference settings of SURGE-D

Setting	Value
Input	16 frames, stride 2, 256×256, ImageNet normalisation
Encoder	V-JEPA 2 ViT-L/16 or ViT-g/16, frozen
Decoder channels	256 (head: 128)
Output	64×64 per frame, central 8 of 16 sampled frames
Loss weights	KL 1, CC 1, NSS 0.1, auxiliary 0.25
Optimiser	AdamW, weight decay 0.01 (prior: 0)
Learning rate (DHF1K)	3×10^{-4} (prior 10^{-3}), warm-up 0.25 epoch, cosine
Learning rate (fine-tuning)	10^{-4} (prior 3×10^{-4})
Epochs, batch size	3, 8
Clips per video and epoch	8 (DHF1K); about 1,500 clips per epoch elsewhere
Augmentation	horizontal flip ($p = 0.5$)
Precision	bfloat16 autocast, gradient clipping at 1.0
Selection	best epoch by holdout CC
Output blur σ	tuned on the holdout over $\{0, 4, 8, 12, 16, 24\}$ pixels
Seeds	0, 1, 2; ensemble = mean of probability maps

4.7. Inference and Post-Processing

A whole video is processed with overlapping windows (Algorithm 4.1). Successive windows advance by $(T/2)\, s = 16$ original frames, so the central sampled positions of consecutive windows tile the video without gaps or overlaps. An original frame that lies between two sampled positions receives the linear interpolation of their maps. The 64×64 map is upsampled bilinearly to the native resolution, smoothed with a Gaussian of standard deviation σ (in native pixels) and normalised to sum to one. The value of σ was chosen on the holdout by CC for each model and each ensemble, and the same map is then used for all metrics, as is common practice in video saliency evaluation. Kümmerer et al. [69] argue instead that each metric should receive the map that is optimal for it under the model's probabilistic prediction; the single CC-tuned map treats all models alike but may understate the scores a model could reach on the other metrics (Section 5.8).

For an ensemble, the members' maps M are averaged before upsampling and blurring. Horizontal-flip test-time augmentation was implemented but not used in any reported result.

Algorithm 4.1 Whole-video inference with SURGE-D

Require: frames $I_0, \ldots, I_{N-1}$; model f; clip length $T = 16$; stride $s = 2$; blur σ
Ensure: saliency maps $\hat{S}_0, \ldots, \hat{S}_{N-1}$ at native resolution

$h \leftarrow (T/2)\, s$; $\quad c \leftarrow (T/4)\, s$; $\quad b \leftarrow -c$ ▷ hop, offset of first central frame, window start
while $b + c < N$ **do**
  $n_k \leftarrow \min(\max(b + ks,\, 0),\, N - 1)$ for $k = 0, \ldots, T - 1$ ▷ clamp at the video ends
  $(S_0, \ldots, S_{T-1}) \leftarrow f(I_{n_0}, \ldots, I_{n_{T-1}})$ ▷ one forward pass, 64×64 maps
  for $m \leftarrow b + c$ **to** $\min(b + c + h,\, N) - 1$ **do**
    $q \leftarrow (m - b)/s$; $\quad j \leftarrow \lfloor q \rfloor$; $\quad w \leftarrow q - j$
    $M \leftarrow (1 - w)\, S_j + w\, S_{j+1}$ ▷ linear interpolation in time
    $\hat{S}_m \leftarrow \mathrm{normalise}\big(\mathrm{Gauss}_\sigma(\mathrm{upsample}_{H \times W}(M))\big)$
  end for
  $b \leftarrow b + h$
end while

4.8. Evaluation Metrics and Statistical Analysis

4.8.1. Metrics

Six metrics were computed, covering the two families distinguished by Bylinskii et al. [17]: distribution-based metrics, which compare the predicted map P with the ground-truth density Q, and location-based metrics, which evaluate P at the fixated pixels of F. All are computed at the native resolution of the ground truth. Let $z(P)$ denote P standardised to zero mean and unit standard deviation over the frame, $\hat{P} = P / \sum P$ and $\hat{Q} = Q / \sum Q$.

- **Correlation coefficient (CC)**, the Pearson correlation between P and Q, $\mathrm{CC} = \frac{1}{|\Omega|} \sum_p z(P)(p)\, z(Q)(p)$, where Ω is the set of pixels.
- **Normalised scanpath saliency (NSS)** [65], the mean standardised saliency at fixated pixels, $\mathrm{NSS} = \frac{1}{\sum_p F(p)} \sum_p F(p)\, z(P)(p)$.
- **Similarity (SIM)**, the histogram intersection $\sum_p \min(\hat{P}(p), \hat{Q}(p))$.
- **Kullback–Leibler divergence (KLD)**, $\sum_p \hat{Q}(p) \log\big(\varepsilon + \hat{Q}(p)/(\hat{P}(p) + \varepsilon)\big)$, with ε the machine epsilon of double precision as in the MIT benchmark code [17]; lower is better.
- **AUC-Judd (AUC-J)** [22], the area under the ROC curve obtained by thresholding P at the saliency values of the fixated pixels, with fixated pixels as positives and all other pixels as negatives. As in the reference implementation, a tiny random jitter breaks ties between quantised values.
- **Shuffled AUC (sAUC)** [21], [66], which takes its negatives from the fixations of other videos instead of from all pixels and therefore does not reward a centre bias. For each video, the fixations of one random frame of each of ten other videos of the same test set are pooled; for each frame, 100 random draws of as many negatives as there are fixations are scored with thresholds in steps of 0.1 and the areas are averaged.

Each metric is averaged over the frames of a video and then over the videos of the test set; on the audio-visual datasets, the dataset score is the mean over its folds. The shuffled AUC depends on the sampling of negatives, so its values are compared only within this thesis.

Validation of the evaluator. The metrics were implemented independently of any existing saliency code. Before any model of this thesis was scored, the evaluator was checked against reference code by re-scoring a fixed set of saliency maps (Section 5.2).

4.8.2. Statistical Analysis

When two models are scored on the same test videos, their difference is assessed with a paired bootstrap over videos [103]. For a metric m and videos $v = 1, \ldots, V$, let $d_v = m_A(v) - m_B(v)$. The mean difference $\bar{d}$ is reported with a 95 % percentile interval obtained from 10,000 resamples of the V videos with replacement. A model is said to be better on a metric only if the interval excludes zero. The bootstrap treats videos as the independent units and therefore accounts for the strong correlation between frames of the same video. On the audio-visual datasets, the test videos of all folds are pooled for this purpose, which is possible because each video is tested exactly once.

Published results of other methods were computed with other evaluation code and, sometimes, other protocol details. They cannot enter a paired test. They are reported side by side with the results of this thesis, and margins below 0.01 CC are treated as ties (Table 4.1).

4.9. Ethical Considerations

The thesis uses only eye-tracking data that were collected and released by other research groups under their own ethical procedures; no human participants were recruited, and no personal data beyond the anonymised gaze recordings and the public video content of these datasets were processed. The licences of the data and models were respected: the Vig gaze data and the V-JEPA 2 checkpoints carry permissive licences, and the benchmark videos were used for research only, as their distributors require.

Saliency prediction has dual-use potential. Models of where people look can improve video compression, accessibility and the evaluation of user interfaces, but they can also be used to design content that captures attention against the viewer's interest. The models of this thesis predict average gaze over groups of observers and contain no information about individuals; the work does not, however, remove the broader risk, and the code accompanying the thesis is intended for research use.

Finally, the experiments consumed electricity on a single workstation graphics card. The design choices that keep the encoder frozen and amortise its cost over many output frames reduce this footprint compared with fully fine-tuned alternatives, and every result was obtained with a fixed number of runs decided in advance rather than by open-ended search.

CHAPTER FIVE

EXPERIMENTS, RESULTS AND DISCUSSION

This chapter reports the experiments in the order in which they were run, because later experiments were planned on the basis of earlier outcomes. After the setup (Section 5.1) and the validation of the evaluation code (Section 5.2), the two preliminary gates are reported: the choice of the video encoder (Section 5.3) and the test of predictive surprise as a fixation cue (Section 5.4). The main results on DHF1K, the transfer to eight further test sets, qualitative examples and the computational cost follow in Section 5.5, and the ablations in Section 5.6. The chapter ends with a discussion organised by research question (Section 5.7) and an assessment of the threats to validity (Section 5.8). Unless stated otherwise, every number in this chapter was produced by the code and evaluator described in Chapter 4, and every test set was scored once per reported model.

5.1. Experimental Setup

All experiments ran on a single workstation (Table 5.1), one job at a time. A monitoring process checked the GPU temperature, hardware events, disk space and memory every minute and could stop training cleanly; it was never triggered.

Table 5.1: Experimental environment

Item	Specification
Graphics Processing Unit (GPU)	NVIDIA RTX A5000, 24 GB
CPU / RAM	Intel Xeon w7-2475X (20 cores) / 256 GB
Operating system	Windows 11 Pro
Language	Python 3.12
Framework	PyTorch 2.14 with CUDA 13.0 [15]
Model loading	Transformers 5.22 [16]
Other libraries	NumPy [104], OpenCV, SciPy

5.1.1. Implementation Details

The settings of Table 4.3 were used throughout. Training one three-epoch DHF1K model took about 15 minutes of training time with the ViT-L encoder (4.7 minutes per epoch of 4,320 clips) and about 31 minutes with ViT-g (9.9 minutes per epoch), plus the holdout evaluation after each epoch. ViT-g fitted in 10 GB of GPU memory at a batch size of eight. In total, the experiments of this chapter comprise 13 DHF1K training runs for the gates, the main program and its ablations, 7 further DHF1K runs for the ViT-g, 600-video and 32-frame experiments,

and 125 fine-tuning runs on the other datasets (including the 16 runs with audio). Holdout lists, configuration files and the log of every decision are part of the code accompanying the thesis.

5.2. Validation of the Evaluator

Because the metrics were implemented from scratch, the evaluator was validated before any SURGE-D model was scored (Table 4.1). The reference maps were the DHF1K validation predictions of SalFoM [10], produced on the same machine with the authors' released weights, for which per-frame scores computed with the evaluation code released by the TMFI-Net authors [83] were available.

The evaluator reproduced the reference scores: CC 0.5639 against 0.5642 ($\Delta = -0.0003$) and NSS 3.3264 against 3.3268 ($\Delta = -0.0004$), well within the tolerance of the rule. The SIM, KLD and AUC-J values of the reference code use other normalisations and could not be compared directly; against a second, independently written evaluator that follows the MIT benchmark conventions, CC, NSS, SIM and KLD agreed per video to within 5×10^{-5} and AUC-J agreed in the mean. Two deviations from the reference procedures were found and corrected during this check, before any model was scored: the first AUC-J implementation lacked the tie-breaking jitter of the reference code (which lowered AUC-J by 0.010 on quantised maps), and the first shuffled AUC pooled all negatives instead of drawing them per split. With the corrected evaluator, SalFoM's validation maps score CC 0.5639, NSS 3.326, SIM 0.4130, KLD 1.297, AUC-J 0.9243 and sAUC 0.7406. These are the SalFoM numbers used in all comparisons below. They are slightly lower than SalFoM's published scores because they are computed on the validation set rather than on the official test set, on which the authors report CC 0.569 and NSS 3.353 [10].

5.3. Selection of the Video Encoder (Gate G1)

The proposal had chosen V-JEPA 2.1 [14], which extends the V-JEPA 2 objective with a loss on the visible tokens and with self-supervision at intermediate layers, specifically to obtain denser features. Its ViT-L model is distilled from a two-billion-parameter teacher and was cooled down at 384×384 pixels [14][1]. Gate G1 compared three ViT-L encoders under identical conditions: V-JEPA 2, V-JEPA 2.1 and UMT-L [48], the encoder family used by SalFoM. Each was frozen and trained with the same decoder for ten epochs with seed 0, all V-JEPA models at 256×256 pixels. UMT-L received 224×224 input because its position table is fixed to that size and to 16 frames.

Table 5.2 shows that V-JEPA 2 led on all four metrics. V-JEPA 2.1 trailed by 0.009 CC, outside the margin of 0.005 within which the proposal's choice would have been kept, so

[1] Checkpoint `vjepa2_1_vitl_dist_vitG_384` from the official repository.

Table 5.2: Gate G1: frozen ViT-L encoders with the same decoder, internal DHF1K holdout (60 videos, every fifth frame, no blur), best of ten epochs, seed 0

Encoder	Best epoch	CC↑	NSS↑	SIM↑	KLD↓
V-JEPA 2 ViT-L (256^2)	3	**0.5782**	**3.314**	**0.4432**	**1.218**
V-JEPA 2.1 ViT-L (256^2)	2	0.5692	3.295	0.4377	1.245
UMT-L (224^2) [48]	2	0.5464	3.101	0.4157	1.323

V-JEPA 2 was adopted for all later experiments. UMT-L trailed by 0.032 CC. With one seed per encoder, part of the gap between the two V-JEPA variants may be seed noise; the gap to UMT-L is several times larger than the seed-to-seed spread observed later (0.003 CC). The ranking of the two V-JEPA variants runs against the expectation that denser features help a dense task. Two explanations are compatible with the data and were not separated: the decoder reads four intermediate layers, which may already supply the local detail that the last layer of V-JEPA 2 lacks, and the V-JEPA 2.1 model was used at a lower resolution than the one at which it was cooled down.

The learning curves in Figure 5.1 were the more consequential outcome of this gate. For every encoder, holdout CC peaked in the second or third epoch and then declined steadily, while training CC kept rising. A frozen encoder with a 7-million-parameter decoder thus overfits the 540 training videos after a few passes. On the basis of these curves, and before any further run, the schedule of the main program was shortened to three epochs (Section 4.6).

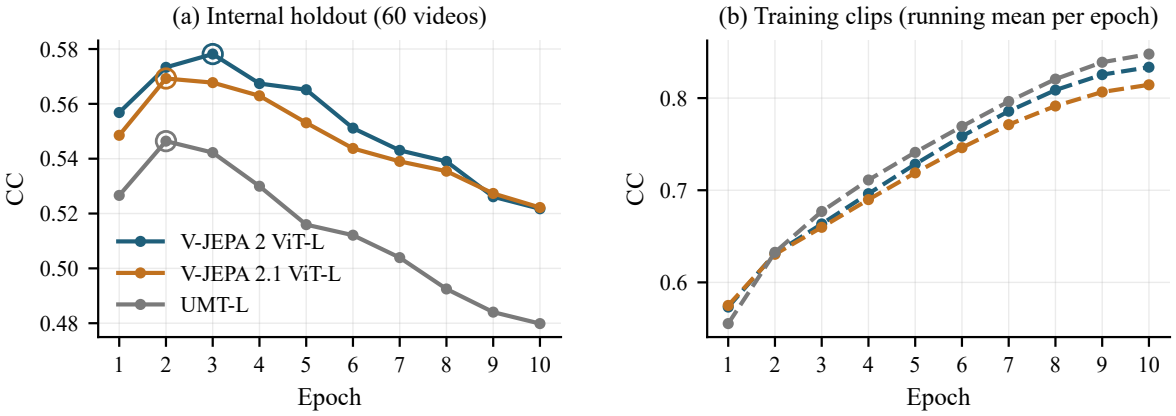


Figure 5.1: Learning curves of gate G1. (a) Holdout CC after each epoch; circles mark the best epoch. (b) Mean CC on the training clips of each epoch. Holdout accuracy peaks early and then declines while training accuracy keeps rising, for all three encoders

5.4. Predictive Surprise as a Fixation Cue (Gate G2)

Gate G2 tested the hypothesis that motivated the original proposal: that the error of V-JEPA 2's pretrained predictor indicates where people look (Section 4.5.1). No training is involved; the surprise maps and the reference maps were scored directly against the fixations of the 60 holdout videos.

Table 5.3: Gate G2: zero-shot fixation prediction by surprise and reference maps on the DHF1K holdout (60 videos), at about 4 and 15 sampled frames per second

	NSS↑		CC↑	
Map	4 fps	15 fps	4 fps	15 fps
Raw surprise, Equation (4.8)	−0.190	−0.502	−0.043	−0.099
Contextual surprise (true − other past)	0.116	−0.366	0.018	−0.072
Token atypicality	0.454	0.650	0.081	0.123
Frame difference	0.810	0.803	0.149	0.148
Empirical centre prior	1.464	1.479	0.289	0.292

The outcome was unambiguous (Table 5.3). Neither surprise variant beat token atypicality or frame difference at either frame rate, and the raw surprise was *anti*-correlated with fixations: its NSS is negative, so observers looked on average at regions that the predictor found *easier* than average to predict. Information gain over the centre prior was negative for every map. The gate therefore failed, and the surprise module was never trained.

A preliminary diagnostic, run before the gate on eleven clips without gaze data (ten Kinetics validation clips [35] and one animated film trailer), helps to explain this result. When the true past of a clip was replaced by another clip's past, the mean prediction error rose in every clip (from 0.644 to 0.731 at about 4 fps), so the predictor does use the context. Yet the spatial pattern of the error correlated at $r \approx 0.60$ with token atypicality and at $r \approx 0.62$ with the error map computed from the *wrong* past, and it correlated negatively with a central Gaussian ($r \approx -0.11$). The error map therefore mainly reflects how unusual each target region is, largely independently of the past, and it is lower at the centre of the frame, where observers tend to look. The released training configuration of V-JEPA 2 offers a plausible reason: its masks span the whole clip in time[2], so the predictor was trained to fill in masked scene content rather than to extrapolate into the future. Its error is consequently not an expectation-violation signal in the sense of Bayesian surprise [13].

5.5. Results

5.5.1. Quantitative Results on DHF1K

The main program trained the frozen-encoder model (labelled B1 in the research log) with three seeds. The LoRA variant lost to it in all three seeds (Section 5.6), so the frozen model became the main model. After the ViT-g encoder passed its own gate, the same recipe was repeated with ViT-g. Table 5.4 reports all four configurations on the DHF1K validation set, next to SalFoM scored by the same evaluator and to published validation results of other methods.

[2] `temporal_scale: [1.0, 1.0]` in the released pre-training configuration files of the official repository.

Table 5.4: Results on the DHF1K validation set (100 videos, 60,273 frames). The upper block was scored by the evaluator of this thesis; the lower block is copied from the literature and was computed with other evaluation code. Best value per column in bold

Model	CC↑	NSS↑	SIM↑	KLD↓	AUC-J↑	sAUC↑
SalFoM [10] (released weights)	0.5639	3.326	0.4130	1.297	0.9243	0.7406
SURGE-D ViT-L, single (σ = 12)	0.5836	3.349	0.4492	1.222	0.9283	0.7636
SURGE-D ViT-L, 3-seed ensemble (σ = 8)	0.5910	3.418	0.4486	1.187	0.9302	0.7614
SURGE-D ViT-g, single (σ = 8)	0.6003	3.457	0.4531	1.170	0.9308	**0.7709**
SURGE-D ViT-g, 3-seed ensemble (σ = 8)	**0.6029**	**3.486**	**0.4574**	**1.160**	**0.9314**	0.7680
Published, other evaluators (as reported in [80])						
TMFI-Net [83]	0.554	3.201	0.428	–	0.924	–
THTD-Net [84]	0.553	3.188	0.425	–	0.924	–
ViNet-E [80]	0.549	3.134	0.409	–	0.922	–
STSANet [81]	0.539	3.082	0.411	–	0.920	–
DiffSal [64]	0.533	3.066	0.405	–	0.918	–

Comparison with SalFoM. SalFoM is the closest prior work: it was the first video saliency model built on a video foundation encoder, it reported the best DHF1K test results among the methods it compared with [10], and its released weights allowed a comparison under an identical protocol. Table 5.5 gives the paired bootstrap intervals. The ViT-g model is better than SalFoM on all six metrics as a single model and as an ensemble, with every interval excluding zero; the CC gain of the single model is +0.036 [+0.026, +0.048]. The ViT-L ensemble is also better on all six metrics, and the single ViT-L model on five, its NSS difference not being significant. The per-video differences in Figure 5.2 show that the gain is broad rather than driven by a few videos: the ViT-g ensemble has the higher CC on 79 of the 100 videos.

Table 5.5: Paired bootstrap differences on DHF1K validation (100 videos, 10,000 resamples), with 95 % intervals; differences whose interval excludes zero are in bold. The intervals for all six metrics are given in Table A.1

Comparison	ΔCC [95 % CI]	ΔNSS [95 % CI]
ViT-L single − SalFoM	**+0.020** [+0.010, +0.031]	+0.023 [−0.050, +0.097]
ViT-L ensemble − SalFoM	**+0.027** [+0.017, +0.038]	**+0.091** [+0.022, +0.164]
ViT-g single − SalFoM	**+0.036** [+0.026, +0.048]	**+0.130** [+0.060, +0.205]
ViT-g ensemble − SalFoM	**+0.039** [+0.029, +0.050]	**+0.160** [+0.089, +0.233]
ViT-g single − ViT-L single	**+0.017** [+0.012, +0.022]	**+0.108** [+0.075, +0.142]
ViT-g ensemble − ViT-L ensemble	**+0.012** [+0.008, +0.016]	**+0.069** [+0.041, +0.097]

Two qualifications apply. First, SURGE-D uses frames on both sides of the predicted frame (Section 4.2), whereas SalFoM predicts the last frame of its input clip; part of the difference may therefore come from access to future context, which SalFoM does not use. Second, the comparison involves one re-scored baseline. The published validation scores in the lower block of Table 5.4 were computed with other code, which agrees with the present evaluator to

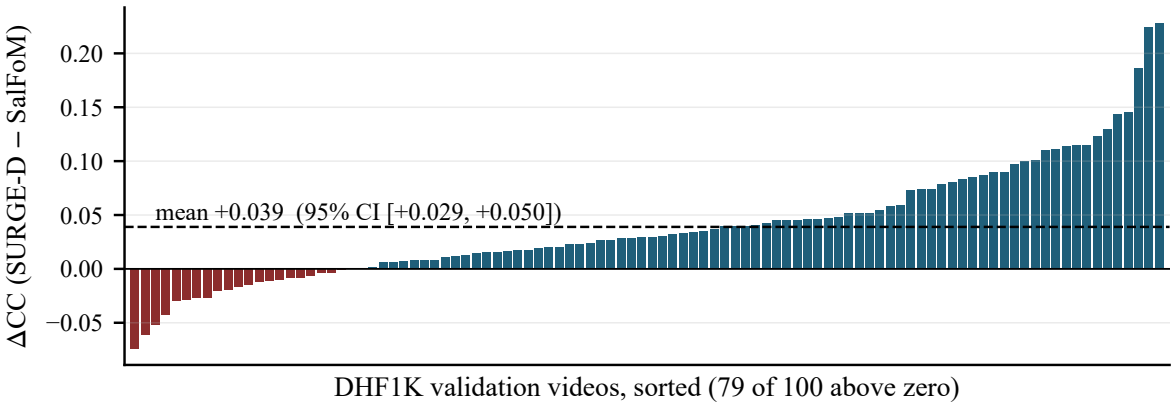


Figure 5.2: Per-video CC difference between the SURGE-D ViT-g ensemble and SalFoM on the 100 DHF1K validation videos, sorted. The dashed line is the mean difference; the interval is the paired bootstrap 95 % interval

within 0.0003 CC on the reference maps (Section 5.2); with that caveat, SURGE-D's validation CC is 0.046–0.049 above the highest of them. Reported validation scores also differ between papers: UniST [85], for instance, lists TMFI-Net at 0.524 rather than 0.554 and reports 0.541 for itself. Several other peer-reviewed models published since 2024 [79], [86], [87] report DHF1K results whose tables could not be accessed or that refer to the test set; test-set scores cannot be compared with the validation scores of this thesis.

DHF1K test set. Maps for the 300 official test videos were produced with the ViT-g three-seed ensemble and checked for format by passing five validation videos through the same export, which reproduced their validation scores to within 10^{-4} CC. The scores on the test set are computed by the dataset authors on request and were not available when this thesis was written, so no test-set result is claimed.

5.5.2. Generalisation to Other Datasets

UCF Sports. UCF Sports differs from DHF1K in two ways: its gaze was recorded during an action-recognition task [56], and it is small (103 training videos). Without any UCF training, the DHF1K model reached a CC of 0.608 (Table 5.6), well below the fine-tuned models, which confirms that the task-driven gaze of this dataset differs from free viewing. After fine-tuning, the ViT-g ensemble reached CC 0.735 and the best NSS, SIM and AUC-J of the SURGE-D models.

On UCF Sports the encoder made a smaller difference than on DHF1K: the ViT-g ensemble was not significantly better than the ViT-L ensemble (ΔCC +0.0019 [−0.0071, +0.0125]), whereas the three-seed ViT-L ensemble was significantly better than its single member (+0.0080 [+0.0039, +0.0123]). Against published results, SURGE-D clearly exceeds SalFoM (by about 0.10 CC) and TMFI-Net, ties ViNet-A (+0.001) and remains below ViNet-E (−0.009, a tie under the rule of Table 4.1, but below ViNet-E on CC, NSS and SIM, though above

Table 5.6: Results on the UCF Sports test set (47 videos, 2,929 frames). Published results were computed with other evaluation code

Model	CC↑	NSS↑	SIM↑	KLD↓	AUC-J↑	sAUC↑
SURGE-D ViT-L, zero-shot	0.6077	3.128	0.5033	1.113	0.9223	0.7453
SURGE-D ViT-L, fine-tuned	0.7252	3.916	0.5799	0.773	0.9406	0.8089
SURGE-D ViT-L, 3-seed ensemble	0.7331	4.028	0.5833	0.749	0.9430	0.8153
SURGE-D ViT-g, fine-tuned	0.7305	4.061	0.5831	0.747	0.9442	0.8145
SURGE-D ViT-g, 3-seed ensemble	0.7350	4.085	0.5857	**0.736**	**0.9449**	**0.8173**
Published ([80]; SalFoM from [10])						
ViNet-E [80]	**0.744**	**4.156**	**0.587**	–	0.941	–
ViNet-A [80]	0.734	4.108	0.586	–	0.940	–
THTD-Net [84]	0.711	3.840	0.565	–	0.933	–
TMFI-Net [83]	0.707	3.863	0.565	–	0.936	–
SalFoM [10]	0.631	3.543	0.516	–	0.928	–

it on AUC-J). SURGE-D is therefore competitive on UCF Sports but not ahead of the best published model. Both ViNet variants use encoders pretrained for action recognition or action localisation [80], which may suit the task-driven gaze of this dataset.

Audio-visual datasets. Table 5.7 reports the six datasets of the STAViS benchmark [63], as fold means. The DHF1K model already transferred reasonably without any training (zero-shot row), and fine-tuning improved every dataset, most of all Coutrot2 (+0.17 CC), whose meetings differ most from DHF1K. With ViT-g, the ensemble was significantly better than the ViT-L ensemble on DIEM, Coutrot1, ETMD and AVAD (paired bootstrap over the pooled test videos), not significantly different on Coutrot2, and tied on SumMe.

Table 5.7: CC on the six audio-visual datasets (fold means; DIEM has one split). The lower block lists the best peer-reviewed results whose tables could be verified, computed with other evaluation code. Visual-only: ViNet-E, or ViNet-A on Coutrot2 [80]; on SumMe, which those models do not report, the visual-only variant of the relevance-guided fusion model AVRSP of Yu et al. [93]. Audio-visual: AVRSP [93] on DIEM and Coutrot2, DiffSal [64] on the other four

Model	DIEM	Coutrot1	Coutrot2	ETMD	AVAD	SumMe
SURGE-D ViT-L, zero-shot	0.675	0.600	0.645	0.588	0.692	0.551
SURGE-D ViT-L, fine-tuned	0.706	0.632	0.816	0.662	0.718	0.581
SURGE-D ViT-L, 3-seed ensemble	0.714	0.641	0.807	0.670	0.729	**0.593**
SURGE-D ViT-g, fine-tuned	0.714	0.648	0.820	0.672	0.731	0.586
SURGE-D ViT-g, 3-seed ensemble	**0.729**	**0.657**	0.819	**0.676**	**0.740**	0.590
Best visual-only (published)	0.701	0.614	**0.862**	0.632	0.729	0.494
Best audio-visual (published)	0.685	0.638	0.837	0.652	0.738	0.572
ViT-g ens. − best visual-only	+0.028	+0.043	−0.043	+0.044	+0.011	+0.096
ViT-g ens. − best audio-visual	+0.044	+0.019	−0.018	+0.024	+0.002	+0.018

Against the best peer-reviewed results that could be verified, the ViT-g ensemble has the higher CC on five of the six datasets. The margin is clear on DIEM, Coutrot1, ETMD and

SumMe. On AVAD, the margin over the best visual-only model is at the threshold (+0.011), and the result ties the best audio-visual model. On Coutrot2 it is lower than both. The other metrics broadly agree with CC: the ensemble's NSS (DIEM 2.88, Coutrot1 3.30, ETMD 3.69, AVAD 4.31, SumMe 3.26) is above the published values of both reference models on these five datasets, while SIM is lower than DiffSal's on Coutrot1 and AVAD.

These comparisons have three limits. They cross evaluation code. They are restricted to peer-reviewed results, so results that had not appeared in peer-reviewed venues when this thesis was written are not included. And they are restricted to results whose tables could be accessed: further recent audio-visual models, among them TM2SP [86], FOSal [105] and the model of Mao et al. [106], could not be checked and are therefore not represented. For AVRSP, as for SalFoM in Table 5.6, the publisher's version could not be accessed, and the numbers were taken from the authors' openly available version of the cited paper. The table should consequently not be read as establishing a state of the art on these datasets. Coutrot2 is discussed in Section 5.7.

Hollywood-2 with the Vig gaze data. Table 5.8 reports the 884 test clips of Hollywood-2 with the gaze of Vig et al. [58] (257,733 scored frames). These numbers are not comparable with published Hollywood-2 results (Section 4.3). With this many test clips, small differences are resolved precisely: ViT-g was better than ViT-L on all six metrics as a single model (ΔCC +0.0046 [+0.0032, +0.0060]) and on five as an ensemble (+0.0041 [+0.0030, +0.0052]). Ensembling raised CC, KLD, AUC-J and sAUC but significantly *lowered* NSS and SIM, for both encoders. With at most five observers per frame, the ground truth is sparse, and the averaging of three members smooths the peaks that NSS and SIM reward.

Table 5.8: Results on Hollywood-2 with the gaze of Vig et al. (884 test clips). Not comparable with published Hollywood-2 results, which use other ground truth

Model	CC↑	NSS↑	SIM↑	KLD↓	AUC-J↑	sAUC↑
SURGE-D ViT-L, zero-shot	0.632	3.595	0.503	1.061	0.951	0.780
SURGE-D ViT-L, fine-tuned	0.6798	3.898	0.5178	0.925	0.9564	0.8167
SURGE-D ViT-L, 3-seed ensemble	0.6841	3.875	0.5073	0.918	0.9574	0.8192
SURGE-D ViT-g, fine-tuned	0.6844	**3.933**	**0.5214**	0.917	0.9569	0.8180
SURGE-D ViT-g, 3-seed ensemble	**0.6882**	3.907	0.5104	**0.911**	**0.9577**	**0.8199**

5.5.3. Qualitative Results

Figure 5.3 shows frames from five DHF1K validation videos, chosen by a rule fixed before any map was inspected: the videos at the 10th, 50th and 90th percentile of the ensemble's per-video CC, and the videos with the largest gain and the largest loss relative to SalFoM; the frame shown is the middle frame of each video.

On these frames, the two models produce similar maps: both locate the horse rider, the



Figure 5.3: Qualitative examples on DHF1K validation. Rows: videos at the 10th, 50th and 90th percentile of SURGE-D's per-video CC, and the videos with the largest gain (0696) and loss (0663) relative to SalFoM; columns: middle frame with the observers' fixations, ground-truth density, SalFoM and the SURGE-D ViT-g ensemble, overlaid on the frame. Numbers give the CC of the shown frame. Video-level CC (SURGE-D / SalFoM): 0650 0.455/0.376, 0632 0.613/0.639, 0685 0.741/0.715, 0696 0.664/0.437, 0663 0.388/0.462

surfer and the flying object that observers follow, and on the single frames shown SalFoM is in fact marginally closer to the ground truth in four of the five rows and equal in the fifth. Single frames are a poor guide to average differences that amount to a few hundredths of CC: for video 0696, for example, SURGE-D's CC over the whole video is 0.23 higher, yet the two maps at its middle frame are almost equal. The last row shows a characteristic failure. In a panel discussion, the observers' fixations concentrate on one person on the right, while SURGE-D spreads its prediction over several faces at the table. In conversation scenes, gaze is drawn to the faces of the people who are speaking [2], which a model that sees only the image cannot determine; the same limitation reappears on Coutrot2 below.

5.5.4. Computational Cost

Table 5.9 reports the cost of SURGE-D, measured with the PyTorch FLOP counter on one $16 \times 256 \times 256$ clip and timed on one DHF1K validation video with frames preloaded into memory (bfloat16, network forward pass only). Because one forward pass yields maps for 16 original frames, the cost per output frame is one sixteenth of the cost per clip.

Table 5.9: Computational cost of SURGE-D on an RTX A5000 (forward pass only). TFLOPs count multiply–add pairs as two operations

	ViT-L	**ViT-g**
Encoder parameters (frozen)	303.9 M	1,012.2 M
Decoder parameters (trained)	6.99 M	7.39 M
Encoder TFLOPs per clip	1.656	5.088
Decoder TFLOPs per clip	0.166	0.168
TFLOPs per output frame	0.114	0.328
Throughput, batch 4 (output frames/s)	301	129
Latency per clip, batch 1 (ms)	116	163
Peak GPU memory, batch 4 (GB)	2.8	7.1

Both models run faster than real time on this GPU. For comparison, the UMT-L encoder that SalFoM fine-tunes requires 1.90 TFLOPs for one $16 \times 224 \times 224$ clip (3,136 tokens; counted on a re-implementation loaded from the released checkpoint), and SalFoM runs it once per output frame in a sliding window [10]. Counting the encoder alone, SalFoM therefore needs about 17 times the computation per output frame of SURGE-D with ViT-L, and about 6 times that of SURGE-D with ViT-g. The multi-frame output thus allows an encoder of three times the size at a fraction of the cost of a per-frame design.

5.6. Ablation Study

Table 5.10 collects the ablations. Each was evaluated on the holdout under the rules of Table 4.1; for single-seed runs, the reference is the frozen ViT-L model of seed 0 trained with the same three-epoch schedule.

Table 5.10: Ablations on the DHF1K holdout (60 videos) unless marked "val". ΔCC is relative to the reference in each block

Variant	CC	ΔCC	Outcome under the rule
Components (ViT-L, seed 0; reference CC 0.5772, NSS 3.329)			
Without centre prior	0.5772	−0.0000	no difference
Without auxiliary heads	0.5757	−0.0015	no difference
With SALICON co-training	0.5801	+0.0028	no difference
Stride 4 (≈7.5 fps, 2 s window)	0.5803	+0.0031	difference (at the threshold)
Encoder adaptation (ViT-L, three seeds)			
LoRA, seed 0 / 1 / 2	0.5743 / 0.5791 / 0.5739	−0.0029 / −0.0010 / −0.0037	rejected (lost 3 of 3)
Encoder scale (seed 0; gate 0.5833)			
ViT-g instead of ViT-L	0.5866	+0.0083 vs ViT-L mean	adopted
Clip length (ViT-g, seed 0; gate 0.5904)			
32 frames instead of 16	0.5902	+0.0036 vs seed 0	not pursued (gate missed by 0.0002)
Training data (ViT-g, val, three-seed ensembles)			
600 instead of 540 train-	0.6044	+0.0015 [−0.0006, +0.0037]	not adopted

Encoder adaptation. LoRA was expected to help, because adapting a pretrained encoder to the target task usually improves on frozen features, and LoRA does so while training only a small fraction of the parameters that full fine-tuning would change [51]. Here it did not. Each LoRA model started from the frozen model of the same seed and lost to it on the holdout in all three seeds (mean −0.0025 CC), and its best epoch was always the first. With only 540 training videos, adapting 1.6 million encoder parameters appears to add more variance than signal; the same early overfitting that limits the decoder (Figure 5.1) would act on the adapted encoder as well. Full fine-tuning, as done by SalFoM, was not tested.

Components. Removing the learned centre prior or the auxiliary heads, and adding 10,000 static SALICON images per epoch, changed holdout CC by less than the threshold of 0.003. The centre prior's lack of effect suggests that the encoder features and the convolutional decoder can already represent the centre bias, for instance through the image borders, which the squashed input keeps at fixed positions; the explicit prior then adds nothing that can be measured. The static images, which helped CNN-era models [8], appear redundant once the encoder has been pretrained on large amounts of video. A stride of four, which halves the cost per output frame, reached +0.0031, exactly at the threshold; with one seed, this is a candidate for future work rather than a finding.

Encoder scale. Replacing ViT-L by ViT-g, with every other setting unchanged, raised holdout CC for all three seeds (0.5866, 0.5867 and 0.5829 against 0.5772, 0.5801 and 0.5776) and validation CC by 0.017 for single models (Table 5.5). Scale was the largest single improvement found after the choice of encoder family.

Temporal context and training data. Doubling the clip to 32 frames (about 2.1 s) reached a holdout CC of 0.5902, 0.0002 below its gate, so it was not pursued; unlike the 16-frame runs, it was still improving at its last epoch, so a longer schedule might change this conclusion, but that would be a new experiment. Training on all 600 DHF1K training videos instead of 540 gave validation scores within noise of the 540-video models. Neither more context nor 11 % more videos made a measurable difference.

Ensembles and output blur. Averaging three seeds raised validation CC by 0.007 for ViT-L and by 0.003 for ViT-g (Table 5.4); the stronger encoder leaves less variance for the ensemble to remove. Figure 5.4 shows that the optimal blur is broad: holdout CC varied by at most 0.003 between $\sigma = 0$ and $\sigma = 12$ for every configuration. The ViT-L ensemble preferred less blur than its single model (8 against 12 pixels), consistent with the smoothing that averaging already provides.

Audio. The audio fusion module of Section 4.5.3 was trained on every fold of the six audio-visual datasets with the same recipe as the visual fine-tuning (ViT-L, seed 0). Table 5.11

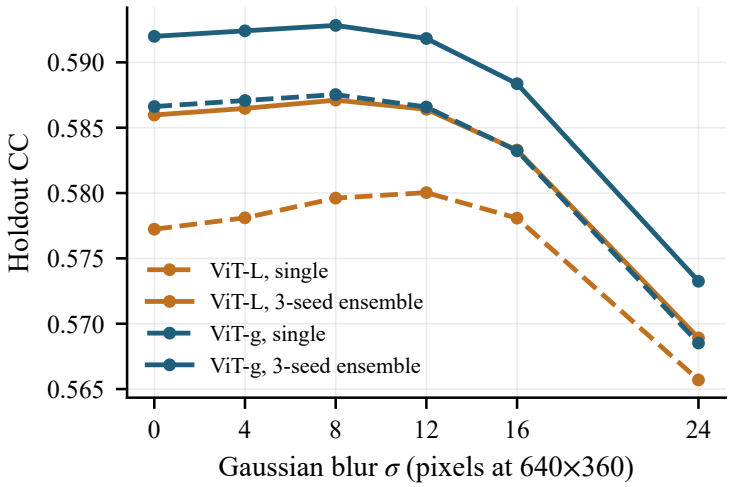


Figure 5.4: Holdout CC as a function of the Gaussian blur applied to the output, for single models and three-seed ensembles. The selected values are 12 (ViT-L single) and 8 (others)

shows that it did not improve the visual model on any dataset: the largest change was +0.001 CC, far below the threshold of +0.010. The control is informative nonetheless. On Coutrot2, scoring the trained model with the wrong soundtrack cost 0.018 CC, so the model had learned to use the audio there, but with the correct soundtrack it only matched the visual model, which already finds the faces in these meeting scenes from the image alone. With five test videos per fold, this observation is suggestive rather than conclusive.

Table 5.11: Audio fusion on the audio-visual datasets (ViT-L, fold-mean CC). The rule requires audio − visual ≥ +0.010 and matched − rotated ≥ +0.005

Dataset	Visual	Audio, matched	Audio, rotated	Audio − visual	Matched − rotated
DIEM	0.706	0.706	0.706	+0.0008	+0.0002
Coutrot1	0.632	0.633	0.630	+0.0010	+0.0031
Coutrot2	0.816	0.816	0.797	−0.0003	+0.0183
ETMD	0.662	0.661	0.661	−0.0014	−0.0001
AVAD	0.718	0.718	0.717	+0.0002	+0.0013
SumMe	0.581	0.579	0.580	−0.0021	−0.0011

5.7. Discussion

This section interprets the results in terms of the research questions of Section 1.4.

RQ1: Can a frozen self-supervised video encoder with a light decoder predict fixations better than the strongest foundation-model approach? On DHF1K, yes, within the limits of the available test data. With ViT-L, a frozen V-JEPA 2 encoder and a 7-million-parameter decoder already exceeded SalFoM, a model whose 300-million-parameter encoder is fully fine-tuned, under an identical evaluation, on five of six metrics as a single model and on all six as an ensemble; with ViT-g, the margin of the single model grew to +0.036 CC and covered

all six metrics. Two features of the result matter for its interpretation. First, the gain does not come from adapting the encoder: the frozen encoder outperformed its LoRA-adapted version, and the result therefore speaks to the quality of the pretrained V-JEPA representations rather than to a training technique. The gate G1 comparison, in which V-JEPA 2 outperformed UMT-L by 0.032 CC under the same frozen protocol, supports this reading, since UMT-L is the encoder family that SalFoM builds on. Second, the evidence is a validation-set comparison with one re-scored baseline and SURGE-D's access to future frames; the official test-set evaluation is still outstanding.

RQ2: Does predictive surprise carry information about where people look? Not in the form available from V-JEPA 2. Its predictor's error was anti-correlated with fixations at 15 fps and was beaten by both trivial reference cues. The diagnostics point to the reason: the error tracks the atypicality of each region rather than violated expectations, and the pre-training masks give the predictor no reason to forecast. This is a statement about one released predictor and one way of reading it out, not a refutation of surprise-based accounts of attention [13]; a predictor trained to extrapolate in time could behave differently.

RQ3: Which design factors matter when the encoder is frozen? The results separate the factors cleanly. The encoder dominates: its pre-training (V-JEPA 2 over V-JEPA 2.1 and UMT-L) and its scale (ViT-g over ViT-L) produced the only gains that were large relative to seed noise on DHF1K. Seed ensembles added a smaller but reliable gain. Nearly everything else, including LoRA, the learned centre prior, deep supervision, static-image co-training, 32-frame clips and 11 % more training videos, made no measurable difference. A consistent explanation is that, with a strong frozen encoder, the limiting factor is the small number of training videos: the decoder overfits within three epochs (Figure 5.1), and the additions that would increase its effective capacity do not help, whereas better features do.

RQ4: How well does the approach transfer, and does audio help? The model transferred well. Fine-tuned for three epochs, it matched or exceeded the best peer-reviewed visual-only and audio-visual results that could be verified on five of the six audio-visual datasets, and it was competitive on UCF Sports, where action-oriented encoders retain an edge. The one clear failure, Coutrot2, has a simple account: in recordings of conversations, observers look at the person who is speaking, which an image-only model cannot determine, as the last row of Figure 5.3 also illustrates. The audio experiment shows that adding the soundtrack in a simple way does not resolve this. The fusion module learned to depend on the audio on Coutrot2, but it did not improve on the visual model. This agrees with the observations of Jain et al. [77] and Agrawal et al. [94], who found that deep audio-visual saliency models tend to become insensitive to their audio input and that none of a range of audio modules and fusion techniques improved on visual-only models across ten datasets. Exploiting sound in this

setting probably requires explicit speaker localisation rather than global feature modulation.

Relation to the original proposal. The final model differs from the proposed one in three respects, each decided by a pre-registered test: the encoder is V-JEPA 2 rather than V-JEPA 2.1, it is frozen rather than adapted with LoRA, and the surprise branch that gave the project its name is absent. The proposal's ranking of what drives performance, which placed the encoder first, was confirmed. Its expectation that LoRA would help, based on results with much larger training sets, was not.

5.8. Threats to Validity

Internal validity. All selection used holdout data, every test set was scored once, and the decision rules were written in advance, with the one exception of the tie margin for published results (Section 4.1); this protects against selection bias. Three residual weaknesses remain. The component ablations used one seed, and they were judged against a fixed threshold rather than a statistical test, so effects of a few thousandths of CC cannot be excluded. The thresholds themselves (for example the margin of 0.005 in the gates) were chosen by judgement, partly before any seed variation had been measured, rather than derived formally. And the tie margin of 0.01 for published results was fixed only after the first such comparison had been seen.

Construct validity. The six metrics capture different aspects of a saliency map [17] and did not always agree; ensembles, for instance, improved CC but lowered NSS and SIM on the sparse Hollywood-2 (Vig gaze) data. The use of one map for all metrics, with the blur chosen by CC, favours CC slightly over the other metrics.

External validity and comparability. Comparisons with published numbers cross evaluation code and, in places, protocol details, such as the handling of frames without fixations or the exact fold composition. Only one baseline, SalFoM, was re-scored under the protocol of this thesis. SURGE-D's predictions use about 0.27 s or more of future context, which causal models do not have. The DHF1K test-set result and the standard Hollywood-2 ground truth were not available. Finally, the pre-training data of V-JEPA 2 include large collections of internet video, and DHF1K is drawn from YouTube; an overlap between pre-training and test videos cannot be ruled out. The same is true for every method that relies on a large pretrained video encoder, including SalFoM.

CHAPTER SIX

CONCLUSION AND FUTURE WORK

6.1. Conclusion

This thesis asked whether the representations of a self-supervised video encoder trained by feature prediction are sufficient for video saliency prediction without adapting the encoder, and whether the encoder's own prediction error indicates where people look. To answer these questions, it developed SURGE-D, which combines a frozen V-JEPA 2 encoder with a light decoder that reads four intermediate layers, rebuilds a multi-scale pyramid, decodes it with factorised spatio-temporal convolutions and predicts saliency maps for sixteen frames per forward pass. Every design decision was tested under a rule written before the corresponding experiment, all models were scored once on their test sets by a single evaluator validated against reference code, and differences were assessed with paired statistics. The answers to the research questions are as follows.

RQ1. A frozen V-JEPA 2 encoder with a light decoder predicted fixations more accurately than SalFoM, the closest foundation-model approach, under an identical protocol. On the DHF1K validation set, the ViT-g model reached a correlation coefficient of 0.600 as a single model and 0.603 as a three-seed ensemble, against 0.564 for SalFoM, and was better on all six metrics with paired bootstrap intervals that exclude zero. This result was obtained without adapting the encoder and at about one sixth of SalFoM's encoder computation per output frame. It is a validation-set result; the official test-set evaluation was pending when the thesis was written, and SURGE-D, unlike SalFoM, uses frames on both sides of the predicted frame.

RQ2. The prediction error of the pretrained V-JEPA 2 predictor did not carry useful information about where people look. It was outperformed by frame difference and token atypicality and was anti-correlated with fixations at 15 frames per second. The error map mainly reflects how atypical each region is and changes little when the true past is replaced by another video's, which is consistent with a predictor trained to fill in masked content rather than to forecast. This finding concerns one released predictor and one way of reading it; it does not refute surprise-based accounts of attention.

RQ3. With a frozen encoder, the encoder itself dominated. Its pre-training (V-JEPA 2 over V-JEPA 2.1 and UMT-L) and its scale (ViT-g over ViT-L) produced the only large gains, and seed ensembles a small one. Low-rank adaptation of the encoder, the learned centre prior, deep supervision, static-image co-training, 32-frame clips and 11 % more training videos

made no difference that could be measured. The early overfitting of even a small decoder suggests that, with a strong encoder, the size of the saliency datasets rather than the capacity of the decoder limits further progress.

RQ4. The approach transferred well. After brief fine-tuning, it was competitive with the best published results on UCF Sports and reached a higher correlation than the best peer-reviewed results that could be verified on five of the six audio-visual datasets; these comparisons cross evaluation code and do not establish a state of the art. Its clear failure was on recordings of conversations, where gaze follows the speaker, which an image-only model cannot identify. Adding the soundtrack through a simple fusion module did not improve any dataset, in line with earlier findings that audio-visual saliency models tend not to exploit the audio.

6.2. Contributions

The thesis makes four contributions. First, it presents SURGE-D, a video saliency model with a frozen self-supervised encoder and a 7-million-parameter multi-frame decoder, which outperformed a fine-tuned foundation-model approach on DHF1K under an identical protocol at a fraction of its computational cost. Second, it reports a pre-registered negative result: the predictor error of a self-supervised video model is not a fixation cue, together with an explanation of why. Third, it provides controlled evidence about which design choices matter once the encoder is frozen, showing that several components commonly included in saliency models add nothing measurable on top of a strong encoder. Fourth, it contributes an evaluation methodology, with a validated evaluator, decision rules fixed in advance, paired statistics and a reproducible ground truth for Hollywood-2 built from openly licensed gaze recordings, that makes its positive and negative findings verifiable.

6.3. Future Work

The limitations of the study point to several directions for future work.

Completing the evaluation. The DHF1K test-set scores of the prepared predictions and an evaluation on the standard Hollywood-2 ground truth would place the results on the benchmarks' official footing. Both depend on the dataset authors and were pending when the thesis was written.

Strengthening the ablations. The component ablations used one seed each. Repeating them with several seeds, and following up the borderline result for a temporal stride of four, which would halve the cost per output frame, would turn the observed absence of effects into tighter bounds. The 32-frame model was still improving at the end of its schedule and merits a longer training run under a new pre-registered rule.

Adapting the stronger encoder. Low-rank adaptation was tested only with ViT-L, and full fine-tuning not at all. Whether adaptation helps the larger ViT-g encoder, or helps once more training data are available, remains open. The V-JEPA 2.1 encoder should also be compared at the resolution at which it was trained.

Speaker-aware audio-visual prediction. The failure on conversations calls for models that determine who is speaking, for example by explicit localisation of the active speaker, rather than modulating visual features with a global audio embedding.

Causal prediction. SURGE-D uses frames on both sides of the predicted frame. A causal variant that uses only past frames would be needed for streaming applications and would allow a stricter comparison with causal models.

Revisiting surprise. The negative result concerns a predictor trained to fill in masked content. Predictors trained explicitly to extrapolate into the future, or surprise measured relative to the scene's own history, may behave differently, and the evaluation protocol of this thesis could test them directly.

Evaluation practice. Scoring each metric with the map that is optimal for it under the model's prediction [69], and reporting paired statistics against re-scored baselines, would make comparisons in the field more informative.

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APPENDIX A

ADDITIONAL RESULTS

This appendix gives the complete statistics behind the summaries of Chapter 5. All values were produced by the evaluator of Section 4.8.

A.1. Paired Bootstrap on the DHF1K Validation Set

Tables A.1 and A.2 extend Table 5.5 to all six metrics. Each cell gives the mean per-video difference and, below it, its 95 % percentile interval over 10,000 bootstrap resamples of the 100 validation videos. For KLD, lower is better, so negative differences favour the first model.

Table A.1: Paired bootstrap differences between SURGE-D and SalFoM on DHF1K validation, all six metrics

Metric	ViT-L single	ViT-L ensemble	ViT-g single	ViT-g ensemble
CC	+0.0197 [+0.0095, +0.0305]	+0.0271 [+0.0171, +0.0376]	+0.0364 [+0.0259, +0.0475]	+0.0390 [+0.0286, +0.0498]
NSS	+0.0225 [−0.0499, +0.0967]	+0.0910 [+0.0218, +0.1637]	+0.1304 [+0.0603, +0.2047]	+0.1597 [+0.0891, +0.2333]
SIM	+0.0362 [+0.0291, +0.0436]	+0.0356 [+0.0285, +0.0430]	+0.0401 [+0.0328, +0.0476]	+0.0444 [+0.0371, +0.0520]
KLD	−0.0747 [−0.1021, −0.0486]	−0.1101 [−0.1359, −0.0858]	−0.1270 [−0.1527, −0.1022]	−0.1369 [−0.1634, −0.1119]
AUC-J	+0.0040 [+0.0022, +0.0060]	+0.0060 [+0.0042, +0.0079]	+0.0065 [+0.0048, +0.0084]	+0.0071 [+0.0053, +0.0090]
sAUC	+0.0230 [+0.0166, +0.0298]	+0.0208 [+0.0143, +0.0277]	+0.0304 [+0.0238, +0.0373]	+0.0275 [+0.0209, +0.0345]

A.2. All Metrics on the Audio-Visual Datasets

Table A.3 reports all six metrics of the SURGE-D ViT-g three-seed ensemble on the six audio-visual datasets, as fold means. The output blur was tuned on each fold's holdout; the selected values lay between 8 and 16 pixels.

A.3. Encoder Comparison on the Audio-Visual Datasets

Table A.4 gives the paired bootstrap comparison of the ViT-g and ViT-L ensembles on the audio-visual datasets. Because every video of these datasets is tested in exactly one fold, the test videos of all folds were pooled for the bootstrap.

Table A.2: Paired bootstrap differences between the ViT-g and ViT-L encoders on DHF1K validation, all six metrics

Metric	Single models	Three-seed ensembles
CC	+0.0167 [+0.0116, +0.0222]	+0.0118 [+0.0077, +0.0163]
NSS	+0.1079 [+0.0746, +0.1423]	+0.0687 [+0.0407, +0.0974]
SIM	+0.0039 [+0.0000, +0.0077]	+0.0088 [+0.0058, +0.0119]
KLD	−0.0523 [−0.0669, −0.0386]	−0.0268 [−0.0381, −0.0158]
AUC-J	+0.0025 [+0.0016, +0.0035]	+0.0011 [+0.0004, +0.0020]
sAUC	+0.0074 [+0.0041, +0.0107]	+0.0067 [+0.0038, +0.0095]

Table A.3: SURGE-D ViT-g three-seed ensemble on the audio-visual datasets (fold means; DIEM has one split)

Dataset	CC↑	NSS↑	SIM↑	KLD↓	AUC-J↑	sAUC↑
DIEM	0.729	2.878	0.584	0.664	0.923	0.735
Coutrot1	0.657	3.302	0.504	0.947	0.911	0.774
Coutrot2	0.819	6.673	0.601	0.683	0.966	0.759
ETMD	0.676	3.687	0.510	0.930	0.950	0.765
AVAD	0.740	4.306	0.565	0.795	0.942	0.782
SumMe	0.590	3.261	0.448	1.148	0.927	0.763

Table A.4: CC difference between the ViT-g and ViT-L three-seed ensembles on the audio-visual datasets (pooled test videos, 10,000 resamples)

Dataset	Test videos	ΔCC [95 % CI]	Reading
DIEM	17	+0.014 [+0.003, +0.027]	ViT-g better
Coutrot1	60	+0.017 [+0.005, +0.031]	ViT-g better
Coutrot2	15	+0.012 [−0.007, +0.032]	no significant difference
ETMD	12	+0.006 [+0.002, +0.010]	ViT-g better
AVAD	45	+0.011 [+0.005, +0.019]	ViT-g better
SumMe	23	−0.003 [−0.010, +0.004]	no significant difference

APPENDIX B

REPRODUCIBILITY AND SAMPLE CODE

The implementation was written for this thesis and does not import code from other saliency projects. This appendix lists its main parts, the commands that produce the main results, and the forward pass of the decoder.

B.1. Organisation of the Code

Table B.1 lists the main files. The research log records every decision rule with its date, written before the experiment it governs, and every result with the folder it was read from.

Table B.1: Main files of the implementation

File	Content
`surge/backbones.py`	frozen encoders (V-JEPA 2 ViT-L and ViT-g, V-JEPA 2.1, UMT-L) and their taps; predictor error for gate G2
`surge/model.py`	feature pyramid, (2+1)D decoder, auxiliary heads, centre prior, surprise and audio modules
`surge/data.py`	dataset readers, clip sampling, targets, inference windows
`surge/losses.py`	training loss (Equation (4.14))
`surge/metrics.py`	CC, NSS, SIM, KLD, AUC-J and sAUC
`surge/infer.py`	whole-video inference (Algorithm 4.1) and scoring
`surge/lora.py`	low-rank adaptation (Equation (4.9))
`scripts/train.py`	training with holdout selection and clean stop/resume
`scripts/evaluate.py`	scoring of models, ensembles or saved maps
`scripts/tune_sigma.py`	selection of the output blur on the holdout
`scripts/compare.py`	paired bootstrap over videos
`scripts/g2_surprise.py`	gate G2
`scripts/prep_hollywood_gaze.py`	Hollywood-2 (Vig gaze) maps
`scripts/prep_stavis_audio.py`	audio features for the audio experiment
`scripts/thesis_qualitative.py`	qualitative examples and cost measurements
`configs/`	configurations and all split and holdout lists
`research/log.md`	dated decision rules and results

B.2. Commands for the Main Results

The main DHF1K model with the ViT-g encoder (seed 0) is trained, its blur is tuned on the holdout, and it is scored once on the validation set with:

```
python scripts/train.py --config configs/main_b1.yaml --set \
    run_dir=runs/vitg_s0 train.seed=0 train.epochs=3 \
```

```
    train.warmup_epochs=0.25 model.backbone=vjepa2_g "data.size=[256,256]"
python scripts/tune_sigma.py --run runs/vitg_s0 \
    --out runs/vitg/sigma_single.json
python scripts/evaluate.py --run runs/vitg_s0 --split val --metrics all \
    --sigma 8 --out runs/vitg/val_single_s0.json
```

Passing `--run` three times (seeds 0, 1 and 2) to the last two commands evaluates the three-seed ensemble. The paired bootstrap against another evaluation file is computed with `python scripts/compare.py A.json B.json`.

B.3. Forward Pass of the Decoder

The listing below is the forward method of the decoder in `surge/model.py`, which implements Equations (4.3), (4.5) and (4.7) (the softmax over pixels is applied in the loss and at inference). The two branches used only by the audio and surprise modules are abbreviated.

```
def forward(self, taps, surprise=None, T_out=None, audio=None):
    p4, p8, p16a, p16b = self.fpn(taps)            # SimpleFPN-3D
    if self.audio:                                  # audio experiment only
        ...
    if self.use_scm:                                # surprise module (not used)
        ...
    d16 = self.r16(self.fuse16(torch.cat([p16a, p16b], 1)))
    d8 = self.r8(p8 + F.interpolate(d16, size=p8.shape[2:], mode="nearest"))
    d4 = self.r4(p4 + F.interpolate(d8, size=p4.shape[2:], mode="nearest"))
    out = {"logits": self.head(d4)[:, 0]}           # B, T', 64, 64
    if self.aux and self.training:
        out["aux16"] = self.aux16(d16)[:, 0]
        out["aux8"] = self.aux8(d8)[:, 0]
    for k in list(out):
        x = out[k]
        if k != "logits":                           # aux maps to 64 x 64
            x = F.interpolate(x[:, None], size=out["logits"].shape[1:],
                              mode="trilinear", align_corners=False)[:, 0]
        if T_out is not None and x.shape[1] != T_out:   # tubelets -> frames
            x = F.interpolate(x[:, None], size=(T_out, *x.shape[2:]),
                              mode="trilinear", align_corners=False)[:, 0]
        if self.prior is not None:                  # + log G_theta
            x = x + self.prior(x.shape[-2], x.shape[-1], x.device)
        out[k] = x
    return out
```